Engineering Structures 366 (2026) 123298

Contents lists available at ScienceDirect

Engineering Structures

journal homepage: www.elsevier.com/locate/engstruct

Digital twin-driven intelligent structural fault diagnosis and decision-making for earth-rock dams using large language models

Jichen Tian [a,b], Chen Chen [a,b,*], Yanling Li [a,b], Huibao Huang [a,b,c], Jiankang Chen [a,b]

[a] State Key Laboratory of Hydraulics and Mountain River Engineering, Sichuan University, Chengdu 610065, China
[b] College of Water Resources & Hydropower, Sichuan University, Chengdu 610065, China
[c] China Yajiang Group Co., Ltd., Beijing 100038, China

ARTICLE INFO

Keywords:
Earth-rock dams
Digital twin
Structural health monitoring
Fault diagnosis

ABSTRACT

Structural health monitoring of high earth-rock dams is essential for ensuring operational safety under complex hydro-mechanical conditions. However, existing digital twin approaches mainly focus on state prediction and lack capabilities for intelligent fault reasoning and engineering decision-making. To address this issue, this study proposes a digital twin-driven intelligent fault diagnosis framework integrated with Large Language Models (LLMs). Multi-source monitoring data, finite element simulations, and transfer learning are combined to construct multi-physics digital twins of deformation, seepage, and stress fields. A robust gradient estimation method and a real-time slope stability model are developed to identify critical failure indicators, including seepage anomalies, cracking risks, hydraulic fracturing, and slope instability. Furthermore, a data semanticization engine is introduced to transform numerical fault information into engineering-oriented natural-language descriptions, which are subsequently analyzed by a dam-specific LLM for root-cause reasoning and mitigation recommendation generation. The proposed framework is validated using a 186 m-high core-wall rockfill dam with long-term monitoring data. The results demonstrate that the method can accurately reconstruct multi-physics fields, effectively identify potential structural risks, and provide interpretable decision-support information, showing strong potential for intelligent safety management of large hydraulic infrastructures.

1. Introduction

Earth-rock dams have been widely constructed and applied worldwide due to their strong adaptability to complex foundation conditions, the possibility of using locally available materials, and their excellent seismic performance [1]. In recent years, China has successively constructed or planned several high earth-rock dams with heights on the order of 300 m, such as Shuangjiangkou, Lianghekou, and Rumei [2,3]. However, during long-term operation, high earth-rock dams are subjected to complex working conditions, including high water pressure, high stress, and particle crushing [4]. Structural failures induced by seepage and deformation still occur from time to time [5–7]. Therefore, carrying out Structural Health Monitoring (SHM) for high earth rock dams is of great engineering significance and can effectively safeguard public safety and property [8,9].

As illustrated in Fig. 1, with the continuous advancement of monitoring technologies, dam operation has generated massive volumes of monitoring data characterized by multiple types and scales [10–12]. In this context, digital twins of high earth-rock dams can effectively predict full-field responses [13,14], such as deformation and seepage fields, thereby enabling multi-source data fusion. However, current SHM methods for high earth-rock dams based on digital twins still face the following challenges: (1) Existing digital twin-based multi-physical-field fault diagnosis approaches remain largely undeveloped. Traditional fault diagnosis methods are limited in their ability to perform real-time and full-field multi-physical-field diagnosis [15]. (2) Fault analysis and response planning still heavily rely on the knowledge and experience of human engineers, lacking systematic approaches for automated fault cause identification and decision-making [16].

To improve the automation of SHM for high earth-rock dams, this study proposes a data-mechanism-knowledge-driven framework based on digital twins and Large Language Models (LLMs). The framework is built on multi-source monitoring data, uses digital twins to achieve full domain mapping of multi-field coupled states, and relies on LLMs to accomplish fault type diagnosis, cause analysis, and mitigation recommendation generation. In this way, interpretable and interactive

* Corresponding author at: State Key Laboratory of Hydraulics and Mountain River Engineering, Sichuan University, Chengdu 610065, China.
E-mail address: laurachen@scu.edu.cn (C. Chen).

https://doi.org/10.1016/j.engstruct.2026.123298
Received 12 April 2026; Received in revised form 29 May 2026; Accepted 28 June 2026
Available online 2 July 2026
0141-0296/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

intelligent monitoring and decision support are realized, significantly reducing manual burden and improving the safety and efficiency of high earth-rock dam operation and management. The proposed framework is a high-level cognitive framework, grounded in physics-based models and expert knowledge, that systematically integrates multi-source data fusion, anomaly detection, and logical reasoning to generate interpretable, real-time diagnostics and decision-support for dam operation without directly controlling physical infrastructure. The main contributions of this study are as follows:

- A hybrid framework for earth-rock dams based on digital twins and LLMs is proposed. It bridges data-driven approaches and knowledge reasoning for the first time, effectively integrates multi-source heterogeneous data, and improves dam management efficiency while reducing reliance on engineers.
- A robust gradient calculation model for twin fields and a high-accuracy real-time slope stability calculation model based on deformation twin fields, seepage twin fields, stress twin fields, and time-varying parameters are proposed, effectively improving the fault diagnosis capability of digital twins for earth-rock dams.
- A data semanticization engine that combines monitoring data and structural mechanisms is proposed, which can effectively transform numerical information into highly abstract natural language and, together with a domain-specific LLM, enables intelligent reasoning, human-machine interaction, and engineering measure generation.

2. Relevant work

Existing SHM methods for high earth-rock dams can be broadly classified into data-driven models and mechanism-driven models [17]. Data-driven approaches, such as statistical models based on the Hydrostatic-Seasonal-Time (HST) theory and various machine learning and deep learning methods, have achieved notable progress in deformation time-series feature extraction [18]. Mechanism-driven approaches rely on finite element models to perform stress, seepage, and stability analyses, often combined with engineering reliability methods to quantify failure risks [19,20]. Recurrent neural networks, including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), are capable of capturing nonlinear temporal dependencies in high-dam deformation [21–23], while Graph Neural Networks (GNN) can model spatial relationships among monitoring points [24,25]. However, these methods are primarily dependent on limited monitoring data and are unable to achieve full-domain state characterization of dams. They are further constrained by cross-domain discrepancies, limited interpretability, and insufficient engineering operability. As a result, their application in identifying complex multi-field coupling behaviors and failure modes of earth-rock dams remains limited.

With the advancement of Industry 4.0 and intelligent infrastructure concepts [26,27], digital twin technology has provided a new paradigm for full-domain perception and multi-field coupling analysis of high earth-rock dams [13,28]. Digital twins establish a dynamic connection between physical and virtual systems through the integration of physical entities, Building Information Modelling (BIM), and Finite Element Method (FEM) simulations, thereby enabling full-domain and multi-physics analysis [29]. In recent studies, BIM models have been increasingly adopted as centralized information carriers for geometry, material properties, monitoring layouts, and operation records, while FEM models provide physically interpretable responses for deformation, seepage, and stress evolution under complex hydro-mechanical conditions [30]. The interoperability between BIM and FEM further improves

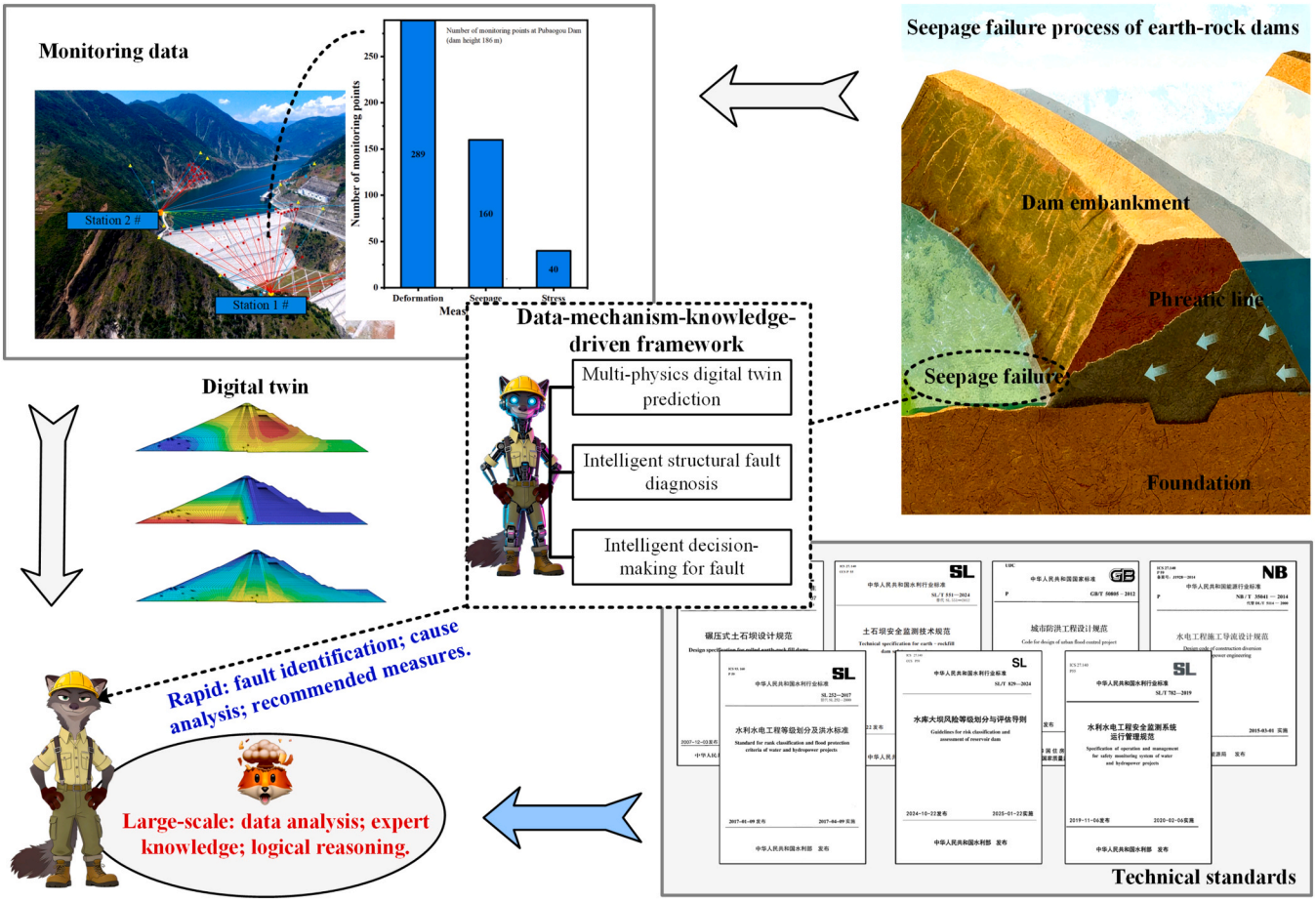

Fig. 1. Functional overview of the data-mechanism-knowledge-driven framework.

data consistency and facilitates the integration of monitoring information with physics-based numerical analysis, thereby enhancing the interpretability and reliability of structural health monitoring systems [31,32].

By integrating physical entities, numerical simulation models, sensor observations, and artificial intelligence algorithms [33,34], digital twins enable synchronized mapping and predictive analysis between physical and virtual systems [35,36]. This paradigm has been validated in structures such as bridges, tunnels, and hydraulic infrastructures [37–40]. Nevertheless, numerical models still rely heavily on extensive calibration of material properties and boundary conditions, and cross-domain discrepancies between numerical simulations and field monitoring data persist, which limits the accuracy of twin fields [14,41]. Moreover, current digital twin frameworks remain insufficient in semantic understanding, fault interpretation, knowledge reasoning, and engineering measure recommendation, and thus cannot yet achieve true intelligent decision making.

Meanwhile, LLMs, with their robust capabilities in natural language understanding and logical reasoning, are rapidly emerging as core technologies for intelligent infrastructure monitoring [42,43]. When fine-tuned with domain-specific knowledge bases, LLMs can transform high-dimensional digital twin outputs into engineering-oriented semantic descriptions, automatically generating fault inferences and mitigation recommendations [44,45]. By incorporating retrieval-augmented generation or knowledge graphs [46], compliance with industry standards and engineering logic is significantly enhanced, enabling digital twins to transition from numerically driven analysis to knowledge-driven reasoning [47–49]. However, a critical review reveals that these emerging digital twin-LLM integrated frameworks are predominantly applied to structures with relatively straightforward boundary conditions, such as standard building frames or logistical site operations [50,51], or treat the LLMs merely as an external natural-language chatbot for database querying, rather than a deeply coupled reasoning engine for multi-physics interpretation [52,53]. For high earth-rock dams, which are characterized by multi-physics and multi-scale coupling, the integration of digital twins and LLMs is still at an early exploratory stage, and systematic framework design and engineering validation remain lacking.

Overall, digital twins provide high-precision virtual representations of multi-field coupled states in dams, while LLMs offer engineering knowledge, natural language reasoning, and human-machine interaction capabilities. How to construct a deeply integrated intelligent agent that unifies data, physical mechanisms, and knowledge to enable intelligent decision making for the structural safety of earth-rock dams remains a critical research gap and constitutes the core problem addressed in this study.

3. Proposed method

To enhance the knowledge, reasoning, automation, and intelligence of existing earth-rock dam digital twin models and to improve the timeliness and effectiveness of structural fault response, this study proposes a hybrid framework integrating digital twins with LLMs. The overall framework consists of three interconnected stages, as illustrated in Fig. 2, where the output of each stage serves as the input for the subsequent analysis and reasoning process. First, a multi-physics digital twin predictive model is developed by combining deep transfer learning with physics-based finite element simulations. By integrating monitoring data, hydraulic conditions, and numerical simulation results, the model reconstructs coupled multi-field states of the dam, including deformation, seepage, and stress fields. The generated multi-physics responses provide both data-driven and mechanically interpretable representations of the structural state, which serve as the foundational input for subsequent fault diagnosis. Second, an intelligent fault

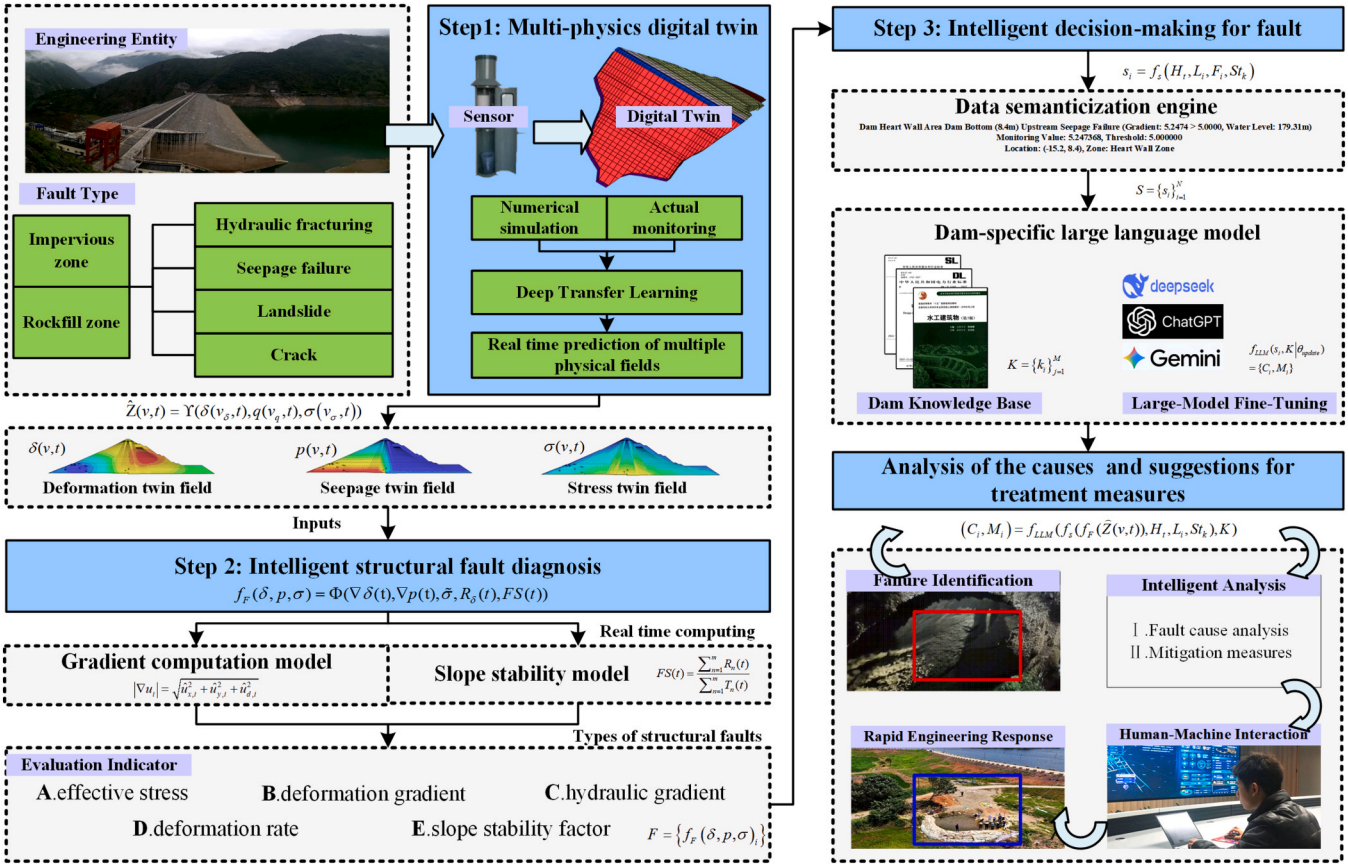

Fig. 2. Workflow of the proposed method.

diagnosis model is established based on the predicted multi-physics states obtained from the digital twin model. Critical safety-related indicators, including deformation rates, deformation gradients, hydraulic gradients, effective stress, and slope stability factors, are extracted and further processed through gradient computation and real-time slope stability analysis. These indicators are subsequently transformed into structural safety feature representations for fault identification, anomaly assessment, and risk characterization. Finally, the identified fault features and safety assessment results, together with contextual engineering information such as structural components, spatial locations, hydraulic loads, and operational conditions, are fed into the LLM-based reasoning module. By incorporating external knowledge bases and task-oriented fine-tuning, the LLM performs knowledge-driven reasoning, root cause analysis, and mitigation strategy generation, thereby enabling intelligent decision support for earth-rock dam safety management.

3.1. Multi-physics digital twin prediction

The multi-physics digital twin relies on the dam sensor network for real-time acquisition of multi-field information, including deformation, seepage, and stress monitoring data, as shown in Fig. 3. In combination with the structural characteristics of earth-rock dams and existing monitoring technologies, the perceptual information is categorized into three types: deformation perception, seepage perception, and stress perception. To further achieve the reconstruction of physical fields from local monitoring points to the entire dam body, this study employs source domain data obtained from finite element numerical simulations and target domain data from sensor monitoring. A digital twin model is constructed by integrating these data with transfer learning techniques.

The model takes multi-source real-time monitoring data as input and outputs the twin fields for deformation, seepage, and stress across the entire dam. The input-output mapping of the multi-field twin prediction model is defined as follows:

$$\hat{\mathrm{Z}}(v,t) = \Upsilon\left(\delta(v_\delta,t), q(v_q,t), \sigma(v_\sigma,t)\right) \tag{1}$$

where $\delta(v_\delta,t)$ denotes the displacement observation vector measured by deformation sensors at spatial position v_δ and at time t, $p(v_p,t)$ represents the monitored seepage data, such as pore pressure, hydraulic head, or flow rate, $\sigma(v_\sigma,t)$ corresponds to the stress monitoring data, including stress or strain measurements. $\Upsilon(\cdot)$ is the multi-field twin mapping

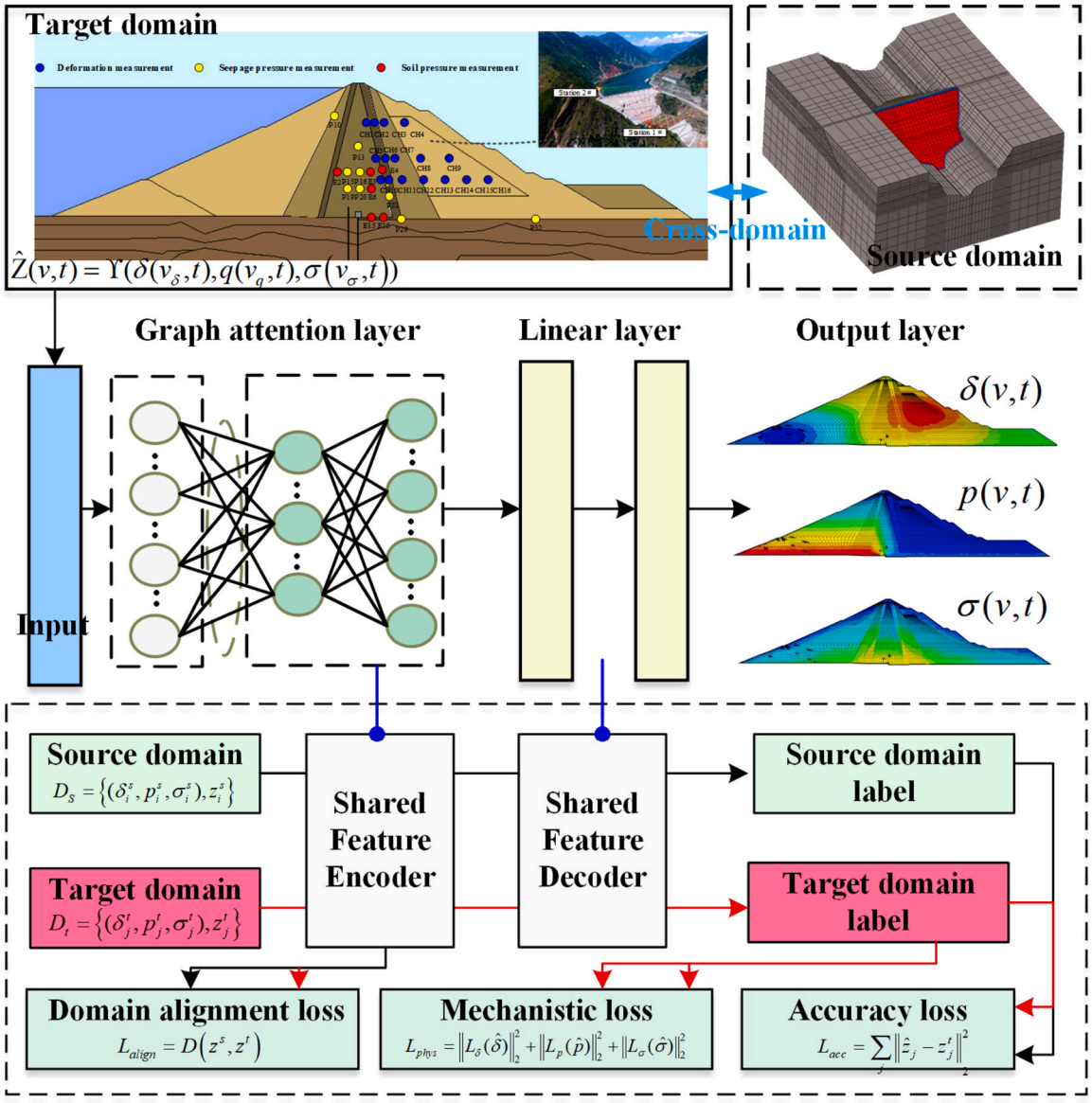

Fig. 3. Workflow of multi-physics digital twin.

function based on Graph Attention Networks (GAT) [14]. $\widehat{Z}(v,t)$ indicates the predicted twin field at spatial position x and time t, comprising the twin deformation field $\delta(v,t)$, the twin seepage field $p(v,t)$, and the twin stress field $\sigma(v,t)$.

During the cross-domain mapping between the source domain (numerical simulations) $D_S = \{(\delta_i^s, p_i^s, \sigma_i^s), z_i^s\}$ and the target domain (monitoring data) $D_T = \left\{(\delta_j^t, p_j^t, \sigma_j^t), z_j^t\right\}$, a joint constraint is imposed through three types of loss functions: accuracy loss, mechanistic loss, and domain alignment loss. The overall optimization objective can be expressed as follows [14]:

$$L = \lambda_1 L_{acc} + \lambda_2 L_{phys} + \lambda_3 L_{align} \tag{2}$$

$$L_{acc} = \sum_j \left\| \widehat{z}_j - z_j^t \right\|_2^2 \tag{3}$$

$$L_{phys} = \|L_\delta(\widehat{\delta})\|_2^2 + \|L_p(\widehat{p})\|_2^2 + \|L_\sigma(\widehat{\sigma})\|_2^2 \tag{4}$$

$$L_{align} = D(z^s, z^t) \tag{5}$$

where L_{acc} denotes the accuracy loss, which ensures consistency between the twin prediction and the target domain monitoring data. L_{phys} denotes the mechanistic loss, which enforces physical consistency by constraining deviations from the governing equations. L_{align} denotes the domain alignment loss, which maintains consistency between the data distributions of the source and target domains. The coefficients λ_1, λ_2, and λ_3 are weighting factors used to balance the relative importance of the different loss terms. $\widehat{z}_j$ represents the predicted physical quantity of the twin model, and z_j^t is the corresponding monitored value. $\|\cdot\|_2$ denotes the Euclidean norm. z^s and z^t represent the feature representations of the source and target domains, respectively, which are used to achieve consistency in the feature space between the two domains.

3.2. Intelligent structural fault diagnosis

The structural fault diagnosis primarily processes the deformation twin field, seepage twin field, and stress twin field transmitted from the multi-physics digital twin and transforms them into corresponding fault types, as shown in Fig. 4. Based on statistical analysis of typical failure types in earth-rock dams and structural mechanistic reasoning, structural faults in earth-rock dams can be classified into two categories: the impervious zone and the rockfill zone. In the impervious zone, the main fault types include seepage failure, uneven deformation, and hydraulic fracturing, which are evaluated using hydraulic gradient, deformation gradient, and effective stress, respectively. In the rockfill zone, the primary fault type concerns slope stability, with evaluation indicators including deformation rate and the factor of safety for slope stability. To achieve high-precision analysis of these structural safety indicators, this study develops real-time gradient computation models based on the deformation twin field, seepage twin field, and stress twin field, as well as a real-time slope stability computation model. These models support accurate identification of fault types and risk assessment, which can be expressed as follows:

$$f_F(\delta, p, \sigma) = \Phi(\nabla\delta(t), \nabla p(t), \widetilde{\sigma}, R_\delta(t), FS(t)) \tag{6}$$

where $f_F(\cdot)$ denotes the fault type identified in the previous stage, ∇ represents the gradient computation results, $\widetilde{\sigma}$ denotes the effective stress, $R_\delta(t)$ denotes the deformation rate, and $FS(t)$ denotes the slope stability factor.

Gradient computation serves as a core procedure for obtaining the local distributions of deformation and seepage fields in the dam.

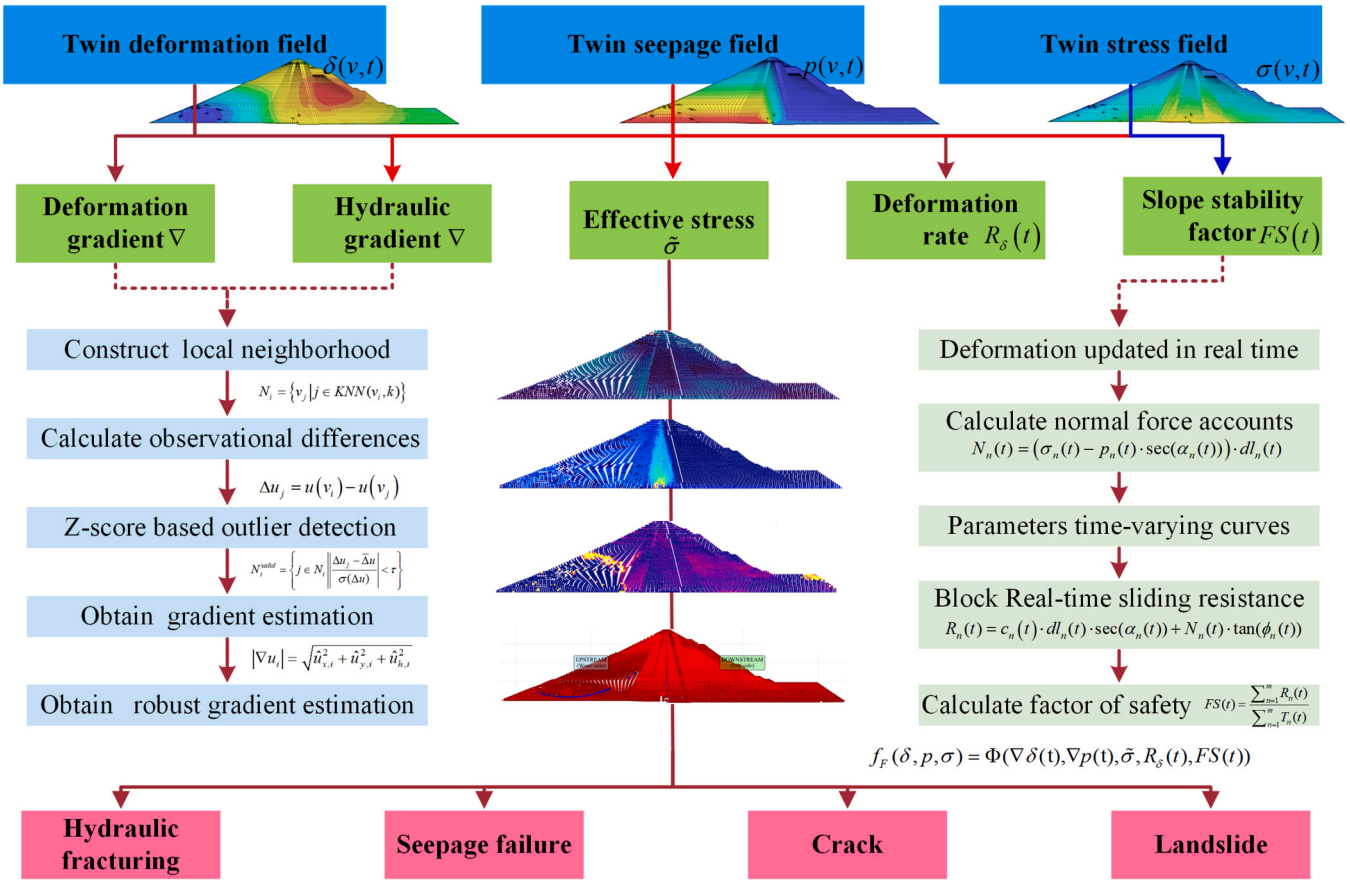

Fig. 4. Workflow of structural fault diagnosis.

Traditional gradient estimation methods based on finite difference or finite element techniques are susceptible to locally abnormal observations and the sparse arrangement of sensors, which can result in unstable gradient estimates. To address this issue, this study proposes a robust gradient estimation method. By incorporating large-neighborhood sampling, outlier detection, and distance-weighted least squares fitting, the method achieves high-precision and robust estimation of displacement gradients and hydraulic head gradients. Let $v_i = (x_i, y_i, h_i)$ denote the spatial coordinates of the i-th monitoring point, and let $u(v_i)$ represent its corresponding twin value, such as deformation or hydraulic head. For each point v_i, its k nearest neighboring points $\{v_j\}$ in three-dimensional space are first selected to construct the local neighborhood N_i:

$$N_i = \{v_j | j \in KNN(v_i, k)\} \tag{7}$$

where $KNN(\cdot)$ denotes the k-nearest neighbor search operator. k provides a balance between estimation stability and locality under typical dam monitoring densities. To enhance robustness against outliers, the observational differences within the neighborhood are expressed as follows:

$$\Delta u_j = u(v_i) - u(v_j) \tag{8}$$

The values are then screened using Z-score based outlier detection, and the twin estimates with absolute Z-scores exceeding the threshold τ are removed:

$$N_i^{valid} = \left\{ j \in N_i \middle| \left| \frac{\Delta u_j - \overline{\Delta u}}{\sigma(\Delta u)} \right| < \tau \right\} \tag{9}$$

where $\overline{\Delta u}$ denotes the mean of the twin increments within the neighborhood, and $\sigma(\Delta u)$ represents their standard deviation. τ is set following the conventional three-sigma criterion, which effectively filters statistically significant outliers while preserving normal structural variability. Within the valid neighborhood N_i^{valid}, a three-dimensional distance-weighted least squares model is employed to obtain the gradient estimation:

$$\min_{\nabla u_i} \sum_{j \in N_i^{valid}} w_j (\Delta u_j - \nabla u_i \cdot (v_j - v_i))^2 \tag{10}$$

where $\nabla u_i = [\partial u/\partial x, \partial u/\partial y, \partial u/\partial d]$ represents the three-dimensional gradient vector at point i, and the weight w_j is inversely proportional to the distance between points. The least squares solution yields the robust gradient estimates $(\hat{u}_{x,i}, \hat{u}_{y,i}, \hat{u}_{h,i})$ and the gradient magnitude:

$$|\nabla u_i| = \sqrt{\hat{u}_{x,i}^2 + \hat{u}_{y,i}^2 + \hat{u}_{h,i}^2} \tag{11}$$

Both the deformation-field gradient and seepage-field gradient are calculated using the proposed gradient estimation model described in Eqs. (7)-(11). The method is applied to different physical fields by assigning the corresponding twin values, such as displacement or hydraulic head, as the input variables for gradient computation.

In the safety evaluation of earth-rock dams, slope stability is a core indicator. Its assessment depends not only on the dam cross-sectional geometry and material properties but also needs to reflect, in real time, the evolution of internal stresses, pore pressures, and local deformation fields. The proposed slope stability indicator model is based on a multi-physics digital twin and the Swedish circular arc method. By continuously updating the real-time geometry of dam blocks, as well as the time-varying pore pressure, compressive stress, and material parameters, the model accurately integrates the driving and resisting forces in sliding mechanics, enabling real-time calculation of the global factor of safety.

Let the dam sliding arc in a two-dimensional cross section be defined by the circle center (x_c, y_c) and radius R, with starting and ending angles θ_s and θ_e. The arc is discretized into m blocks. The original midpoint coordinates of each block are (x_n^0, y_n^0), and their positions are updated in real time according to the digital twin deformation field as follows:

$$x_n = x_n^0 + \Delta x_n(t), y_n = y_n^0 + \Delta y_n(t) \tag{12}$$

where $\Delta x_n(t)$ and $\Delta y_n(t)$ represent the lateral and vertical displacements at the midpoint of each block in the digital twin deformation field, calculated in real time over time t. The block height $h_n(t)$, length $dl_n(t)$, and toe position $\alpha_n(t)$ are updated in real time. The normal force accounts $N_n(t)$ for the real-time effects of the digital twin stress field $\sigma(v,t)$ and the seepage pore pressure $p(v,t)$:

$$N_n(t) = (\sigma_n(t) - p_n(t) \cdot \sec(\alpha_n(t))) \cdot dl_n(t) \tag{13}$$

where the friction angle $\phi_n(t)$ and cohesion $c_n(t)$ are generated as time-varying curves by parameter inversion fitting of historical and real-time data. Unlike traditional static methods, which assume constant material properties based on design or laboratory values, this approach captures the time-dependent behavior of high earth-rock dam materials, such as particle crushing and consolidation. Modelling these parameters as time-varying improves the accuracy and timeliness of stability assessments under changing operational conditions. The real-time sliding resistance $R_n(t)$ of each block is expressed as:

$$R_n(t) = c_n(t) \cdot dl_n(t) \cdot \sec(\alpha_n(t)) + N_n(t) \cdot \tan(\phi_n(t)) \tag{14}$$

The overall factor of safety $FS(t)$ satisfies the iterative equilibrium condition:

$$FS(t) = \frac{\sum_{n=1}^{m} R_n(t)}{\sum_{n=1}^{m} T_n(t)} \tag{15}$$

where $T_n(t)$ represents the driving force along the slope direction.

3.3. Intelligent decision-making for fault

The intelligent decision-making module processes fault information $F = \{f_F(\delta, p, \sigma)_i\}$ transmitted from the structural fault diagnosis stage and converts it into structured semantic representations, as shown in Fig. 5. Based on a dam-specific LLM, this process enables knowledge-driven reasoning to generate targeted engineering recommendations, thereby facilitating effective interaction between engineers and the digital twin-based diagnostic system. The data semanticization engine serves as a critical bridging component within the proposed framework. Its primary role is to map heterogeneous fault-related information, including structural failure types, hydraulic loads, spatial coordinates, and dam segments, into structured variables that are explicitly associated with physically meaningful entities of the dam system. In this sense, it is not a simple text conversion module, but a constrained semantic mapping process. The generated variables are required to satisfy predefined physical and engineering consistency constraints, ensuring alignment with structural mechanics interpretations and monitoring system definitions. Only such standardized and physically interpretable representations are passed to the subsequent reasoning stage. In this study, the LLM module is not used as a general-purpose conversational chatbot. Instead, it is configured as a task-oriented reasoning engine that accepts structured semantic inputs and outputs constrained engineering interpretations, including fault diagnosis explanations and corresponding mitigation suggestions. The model operates on structured inputs rather than free-form dialogue, ensuring consistency with engineering semantics. Overall, the data semanticization engine ensures that multi-source monitoring data are consistently anchored to the dam's physical structure, enabling reliable downstream reasoning and decision-making.

By considering hydraulic load, spatial position, dam segments, and fault type, it maps numerical anomalies into readable and inferable semantic descriptions, such as "seepage anomaly at the base of the core

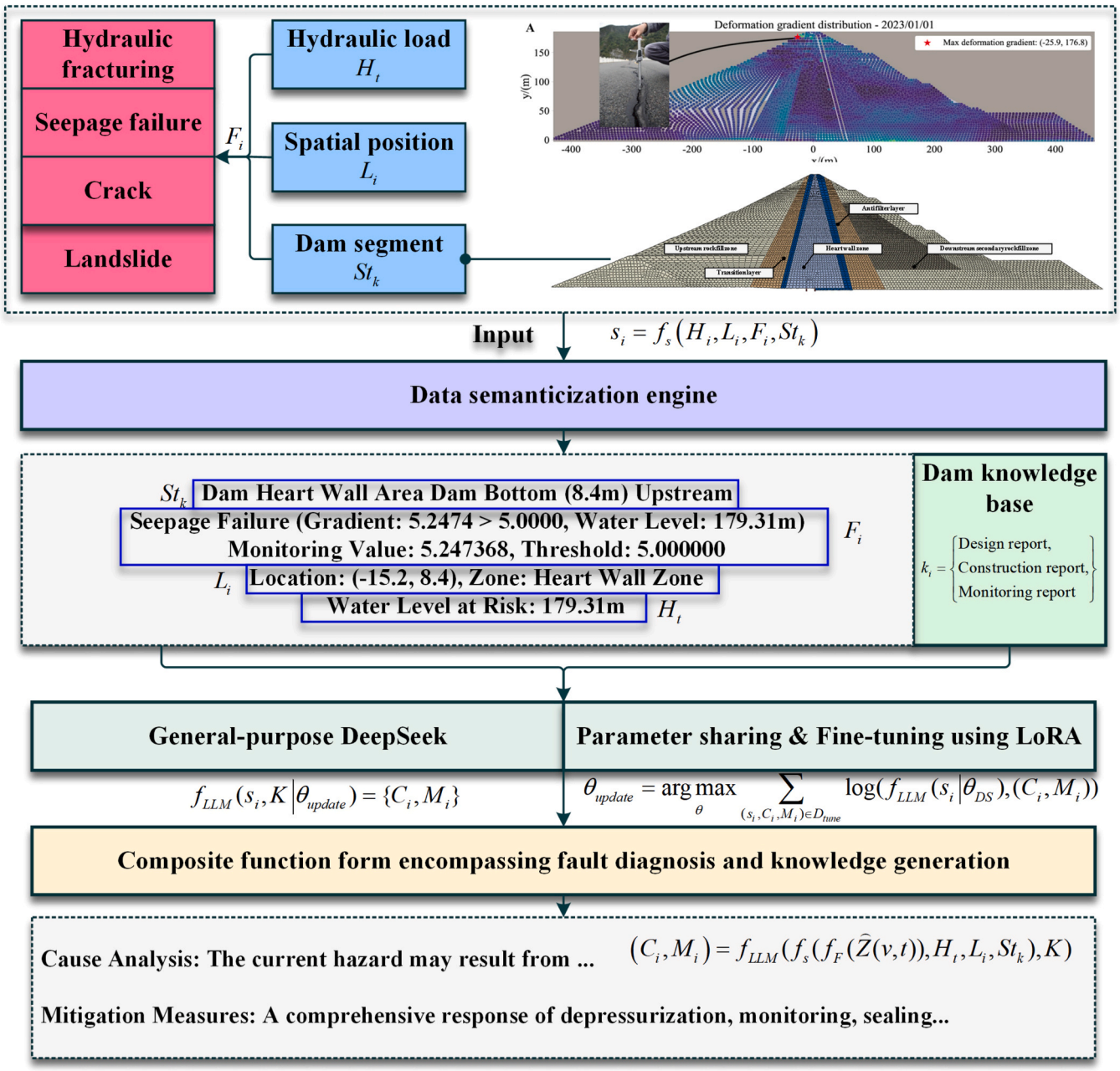

Fig. 5. Workflow of intelligent decision-making.

wall" or "slope displacement anomaly in the downstream rockfill zone", as shown in Fig. 5. The data semanticization engine can be defined as the following mapping:

$$s_i = f_s(H_t, L_i, F_i, St_k) \tag{16}$$

where $S = \{s_i\}_{i=1}^{N}$ represents the high-level semantic output corresponding to each twin node, St_k indicates specific dam segments such as the core wall, transition zone, or rockfill zone. H_t represents hydraulic load. L_i denotes the spatial position of the monitoring point, and F_i indicates the fault type, such as hydraulic fracturing, seepage failure, cracking, or landslide.

The dam-specific LLM serves as the core intelligent unit for fault type identification, root cause analysis, and mitigation strategy generation. This model consists of two main components: a domain knowledge base and a Low-Rank Adaptation (LoRA)-based fine-tuning module. The knowledge base provides structured domain expertise, while the LoRA-based fine-tuning ensures consistency in the input-output formats and improves the model's ability to generate task-specific responses. At the knowledge construction level, the LLM relies on a systematically integrated, dedicated dam knowledge base K, including engineering design, construction, and monitoring reports. In addition to project-specific documentation from the case study dam, the knowledge base also incorporates authoritative industry standards, including Specifications for Rolled Earth-Rockfill Dam Construction (SL/T 848–2025), Design Specification for Rolled Earth-Rock Fill Dams (NB/T 10872–2021), and Technical Specification for Earth-Rock Fill Dam Safety Monitoring (SL/T 551–2024). Formally, this can be expressed as:

$$K = \{k_i\}_{j=1}^{M}, k_i = \{\text{Designreport}, \text{Constructionreport}, \text{Monitoringreport}\} \tag{17}$$

In terms of model construction, this study builds a dam-specific LLM based on the general-purpose DeepSeek, employing a parameter sharing

θ_{DS} and fine-tuning strategy using LoRA [44,45]. The fine-tuning dataset D_{tune} consists of a high-quality dam safety monitoring corpus, including abnormal event handling records and a collection of domain-specific terminology. During actual inference, given an input semantic query $S = \{s_i\}_{i=1}^{N}$, the LLM performs real-time retrieval from the external knowledge base K to identify the most relevant standard entries and case excerpts. The inference process can be formally represented by the following mapping function:

$$f_{LLM}(s_i, K|\theta_{update}) = \{C_i, M_i\} \tag{18}$$

$$\theta_{update} = \arg\max_{\theta} \sum_{(s_i, C_i, M_i) \in D_{tune}} \log(f_{LLM}(s_i|\theta_{DS}), (C_i, M_i)) \tag{19}$$

where θ_{update} denotes the optimized parameters obtained via domain-specific fine-tuning, C_i is the fault cause analysis and M_i is the mitigation measures. Furthermore, the inference process can be elaborated as a composite function form encompassing fault diagnosis and knowledge generation:

$$(C_i, M_i) = f_{LLM}\left(f_s\left(f_F\left(\widehat{Z}\left(v, t\right)\right), H_t, L_i, St_k\right), K\right) \tag{20}$$

3.4. Case study

To validate the effectiveness of the proposed method, this study selects a high-core-wall rockfill dam with comprehensive monitoring data as the research subject. The dam is located in the Dadu River basin and is a large-scale hydropower project primarily used for power generation, while also serving flood control, sediment retention, and other integrated functions.

3.5. Project description

The PB Dam is a gravel-core rockfill dam with a maximum structural height of 186 m, defined as the elevation difference between the foundation level (670 m) and the dam crest (856 m). The normal reservoir water level is 850 m, the flood-season operating limit is 841 m, and the dead water level is 790 m, resulting in a maximum drawdown depth of 60 m. The total reservoir capacity is 5.337 billion m^3 , and it functions as a partially regulated reservoir. As shown in Fig. 6**A**, the layered displacements of the downstream transition material and the upstream and downstream directions of the rockfill body are monitored using horizontal extensometers. In the cross-section at chainage 0 + 240 m, sixteen monitoring points are arranged at elevations of 808 m, 758 m, and 731 m, numbered CH1 to CH16. Piezometers P10 to P32 are installed at the junctions between the core wall, downstream dam body,

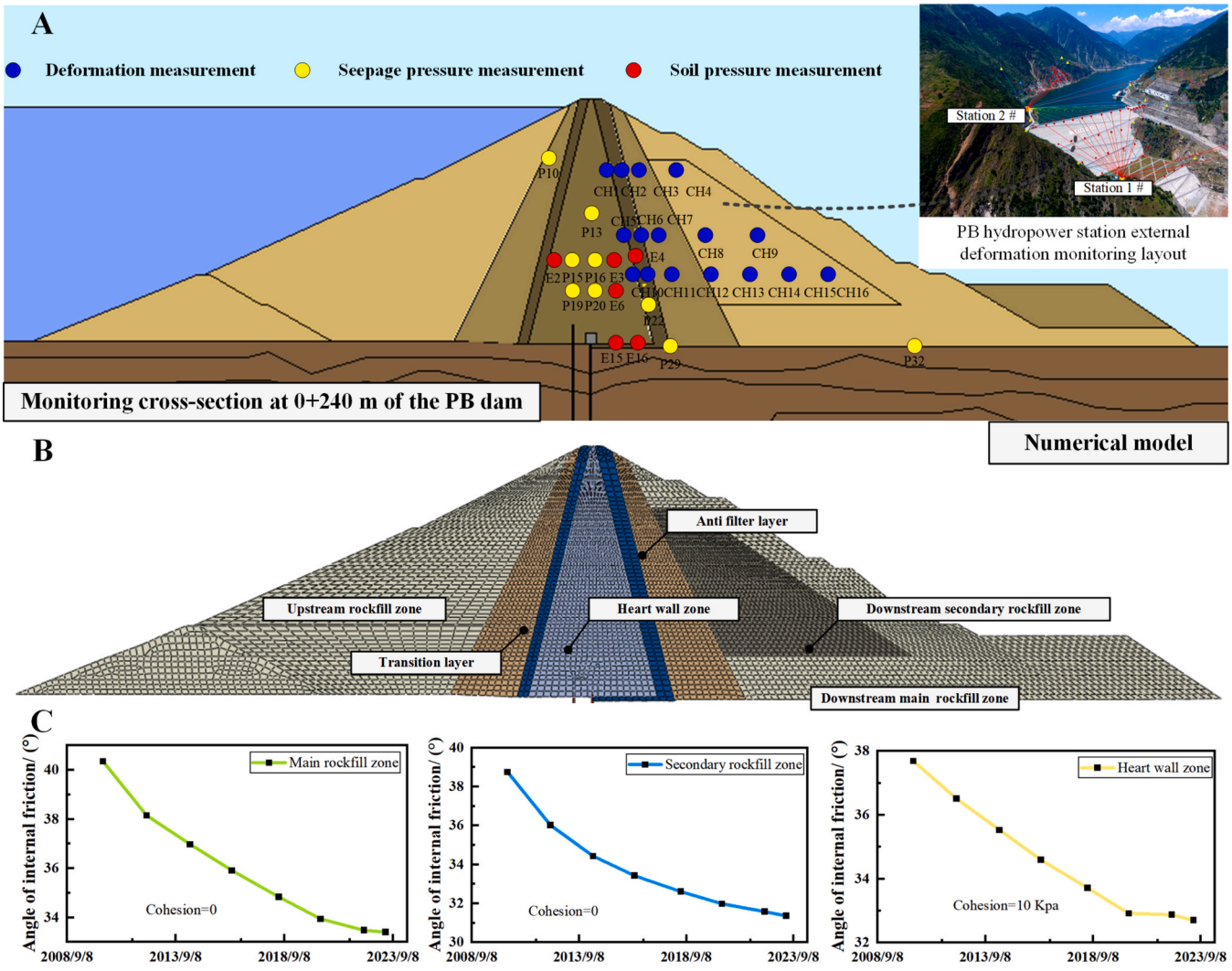

Fig. 6. Monitoring point arrangements for deformation, seepage, and stress at the PB Dam, material zoning of the numerical model, and the time-varying curves of material parameters.

and foundation to monitor pore pressure, while six earth pressure cells, E4 to E16, are installed at the core wall to support the construction of the PB Dam digital twin model.

A numerical model is established using the finite element method in ABAQUS. A two-dimensional plane strain model is adopted based on the dam design drawings, with material zoning implemented as shown in Fig. 6**B**. The geometry of the model is directly derived from engineering design documents, ensuring consistency with the actual dam configuration. The simulation domain extends to twice the dam height in the upstream, downstream, and foundation directions to minimize boundary effects. Appropriate boundary conditions are applied, including prescribed hydraulic head boundaries at the upstream and downstream reservoirs and flux-controlled seepage boundaries at the foundation. The finite element mesh consists of 6572 elements and 13,348 nodes. Material parameters [13] are determined based on design values and further refined using time-varying parameter inversion, and multiple parameter sets are used to construct the source-domain simulation dataset.

The transient deformation behavior is modeled using the Duncan-Chang constitutive model, while time-dependent creep effects are represented using a seven-parameter creep model. The creep model captures the macroscopic time-dependent deformation caused by particle rearrangement and crushing, although it does not explicitly describe the progressive degradation of strength-related parameters. To account for this limitation, the internal friction angles of the zoned materials used in the slope stability analysis are represented as time-varying parameters through period-based parameter inversion [54,55]. In this process, the parameter evolution is fitted using an appropriate degradation function, such as an exponential function, according to the historical inversion results and monitoring trends. This approach does not involve real-time parameter adjustment of the numerical model, but rather provides a formalized description of long-term material degradation. As a result, the model can better reflect irreversible processes such as particle crushing and consolidation, thereby improving its ability to capture long-term slope stability evolution, as shown in Fig. 6**C**.

3.6. Dataset description

Based on the actual monitoring configuration of deformation, seepage, and stress at the PB Dam (as summarized in Table 1), this study constructed the target-domain dataset using the monitoring records from extensometers, piezometers, and earth pressure cells collected between 28 May 2012 and 23 May 2024. The measured data from various monitoring instruments were used as inputs, while the digital-twin predictions of the full-domain physical fields of the PB Dam were used as outputs. The dataset was randomly divided into training, validation, and test sets with a ratio of 0.7:0.1:0.2. Meanwhile, a series of operating-condition simulations was generated using the established finite element model, incorporating both design parameters and parameter-inversion results. These simulations formed the source-domain dataset, which was employed to enhance the model's generalization capability and improve the reconstruction accuracy of the twin fields. The dam-specific LLM was fine-tuned based on DeepSeek-V2.5. A total of 108 high-quality samples were used for the fine-tuning process. The input to the model is a semantic query $S = \{s_i\}_{i=1}^{N}$, while the output includes fault cause analysis C_i and corresponding mitigation measures M_i. In addition, the GAT was employed in the construction of the multi-physics digital twin model. The hidden layer dimension of the GAT was set to 192, and the attention head was set to 4. Based on the finite element mesh density, the number of nearest neighboring points k used in the gradient calculation was set to 20, and the Z-score threshold τ was set to 2.5.

4. Results and analysis

The experiments were conducted in a Windows 10 operating system environment using the PyCharm IDE and the PyTorch 2.5.1 platform based on Python 3.10. The server hardware configuration included an Intel Core i7–13700 CPU and an NVIDIA RTX 4070 SUPER GPU. During the digital twin modelling process, the learning rate was set to 0.001, the batch size was 64, and the Adam optimizer was used to update the network parameters. The training was performed for 500 iterations. To quantitatively evaluate the accuracy and performance of the digital twin model, three metrics were selected: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) [56].

4.1. Accuracy of the digital twin model

To evaluate the accuracy of the proposed method in digital twin modelling, the test set from December 31, 2021, to May 23, 2024, was selected to systematically compare the predicted deformation, seepage, and stress twin fields with the corresponding measured monitoring data. By examining the consistency with physical mechanisms, the variation trends of monitoring time histories, and quantitative indicators such as MAE, MAPE, and RMSE, the model accuracy was thoroughly assessed. Fig. 7**A** presents the deformation, seepage, and stress twin fields of PB Dam at the time instant of December 31, 2021. The results indicate that all three twin fields accurately capture the actual deformation patterns, hydraulic gradient distribution, and stress transfer behavior, and their spatial characteristics are consistent with the theoretical trends of numerical simulation. This demonstrates that the model can effectively reconstruct the internal physical evolution of the dam. Fig. 7**B** compares the predicted twin curves with the measured time histories at selected monitoring points. It can be observed that the deformation prediction of CH4 matches the measured data well; although other deformation points, such as CH1, exhibit certain deviations, their overall trends remain consistent, and the discrepancies mainly arise from cross-domain differences. Nevertheless, the prediction accuracy is significantly improved compared with traditional numerical simulation methods [28, 13,14]. The fitting performance of seepage and stress points is generally good, with the variation trends closely following the measured data, except for slight deviations near E16.

Fig. 7**C** presents the statistical results of MAE, RMSE, and MAPE for all monitoring points in the test set. The average MAE values for deformation, seepage, and stress points are 6.63 mm, 1.88 m, and 13.72 kPa, respectively. Since the pore pressure is close to zero at certain times, the MAPE of seepage points is not representative; however, the average MAPE of deformation and stress points reaches 1.12% and 1.80%, indicating low overall prediction errors. In summary, the digital twin model developed using the proposed method exhibits high accuracy and can meet the requirements for real-time full-domain structural safety assessment of dams.

Table 1
Datasets specifics table.

Datasets	Period	Length	Input dimension			Output dimension
			Deformation	Seepage	Stress	
Source Domain/ Target Domain	2012–2024	4372	16	9	6	6470 × 3

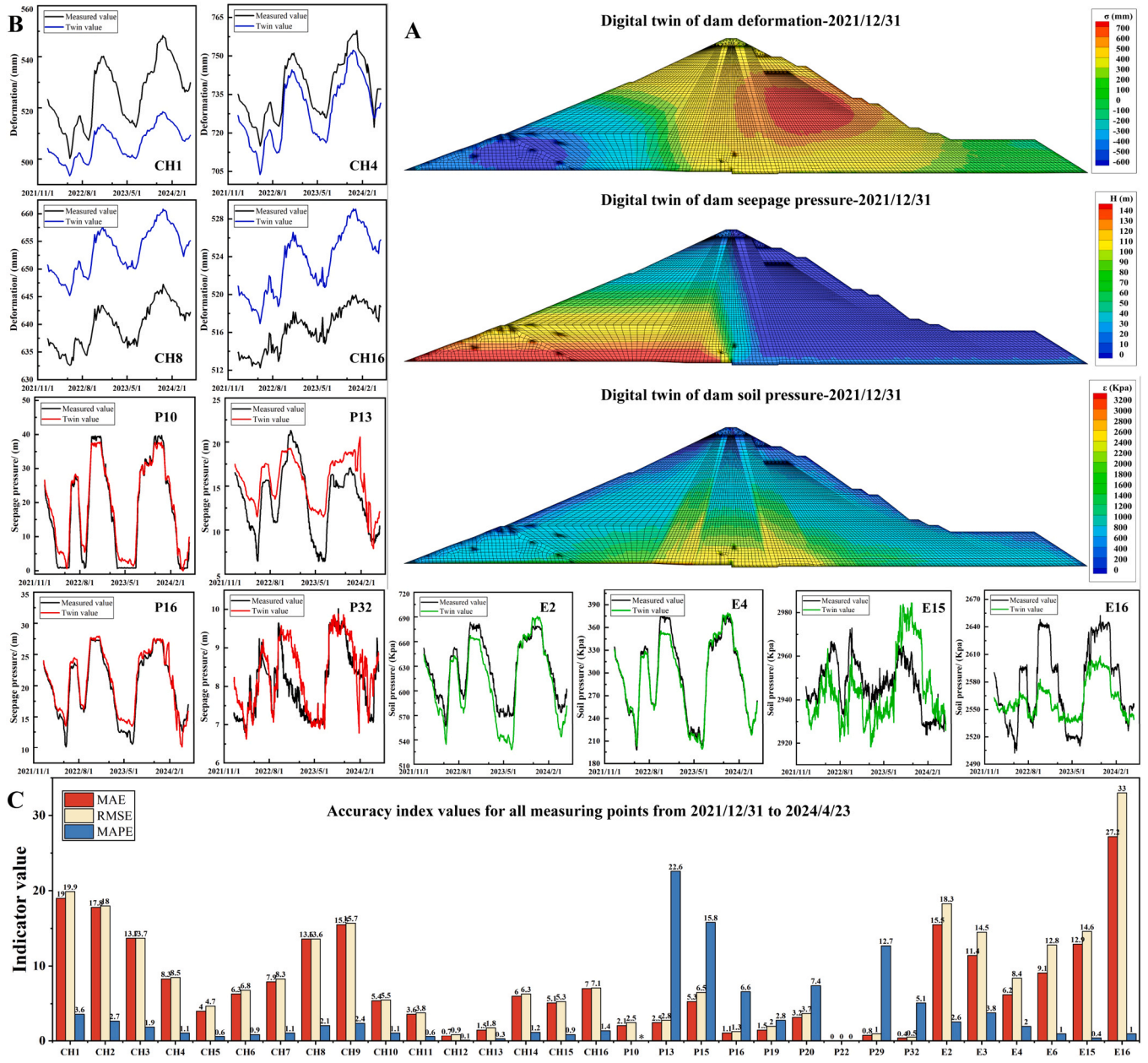

Fig. 7. Accuracy analysis of the proposed digital twin model.

4.2. Fault diagnosis results

To evaluate the effectiveness of the fault diagnosis in the proposed method for high earth-rock dams based on deformation, seepage, and stress twin fields, a systematic examination was conducted on all twin prediction points by visualizing key indicators, including hydraulic gradient, deformation gradient, effective stress, deformation rate, and slope stability. In addition, the rationality of the fault diagnosis results was verified by considering the spatial distribution of the most critical zones, their temporal evolution, and coupling with water levels. As shown in Fig. 8 to Fig. 13, the spatial distribution of all indicators across the full domain of the PB Dam digital twin from January to November 2023 is presented, along with their dynamic evolution from December 31, 2021, to May 23, 2024.

As shown in Fig. 8, the evaluation results of the hydraulic gradient in the PB Dam digital twin are presented. The results indicate that the hydraulic gradient in the rockfill zones is approximately zero, while the core wall experiences relatively high seepage pressure, around 2.0, with the maximum values occurring at the junction between the concrete cutoff wall and the core wall (Fig. 8**C**). The downstream rockfill zone exhibits an almost negligible phreatic line, with seepage occurring only at the interface with the dam foundation. Fig. 8**B** shows the temporal variation of the maximum pore pressure, which demonstrates a certain coupling with the water level fluctuations. The maximum value reaches approximately 5.5, remaining within the allowable design hydraulic gradient range of the cutoff wall and core wall. The observed patterns are consistent with the seepage behavior of high-core rockfill dams and adequately reflect the dam's seepage failure assessment.

As shown in Fig. 9, the evaluation results of the deformation gradient in the PB Dam digital twin are presented. The results indicate that the deformation gradient in the upstream rockfill zone is approximately 1.5%, while the core wall exhibits relatively minor differential deformation, which may be influenced by seepage pressure. The crest shows larger differential deformation, with the maximum values concentrated near the dam top (Fig. 9**C**). Considering the allowable limits of PB Dam, the predicted longitudinal cracking at the crest is highly consistent with

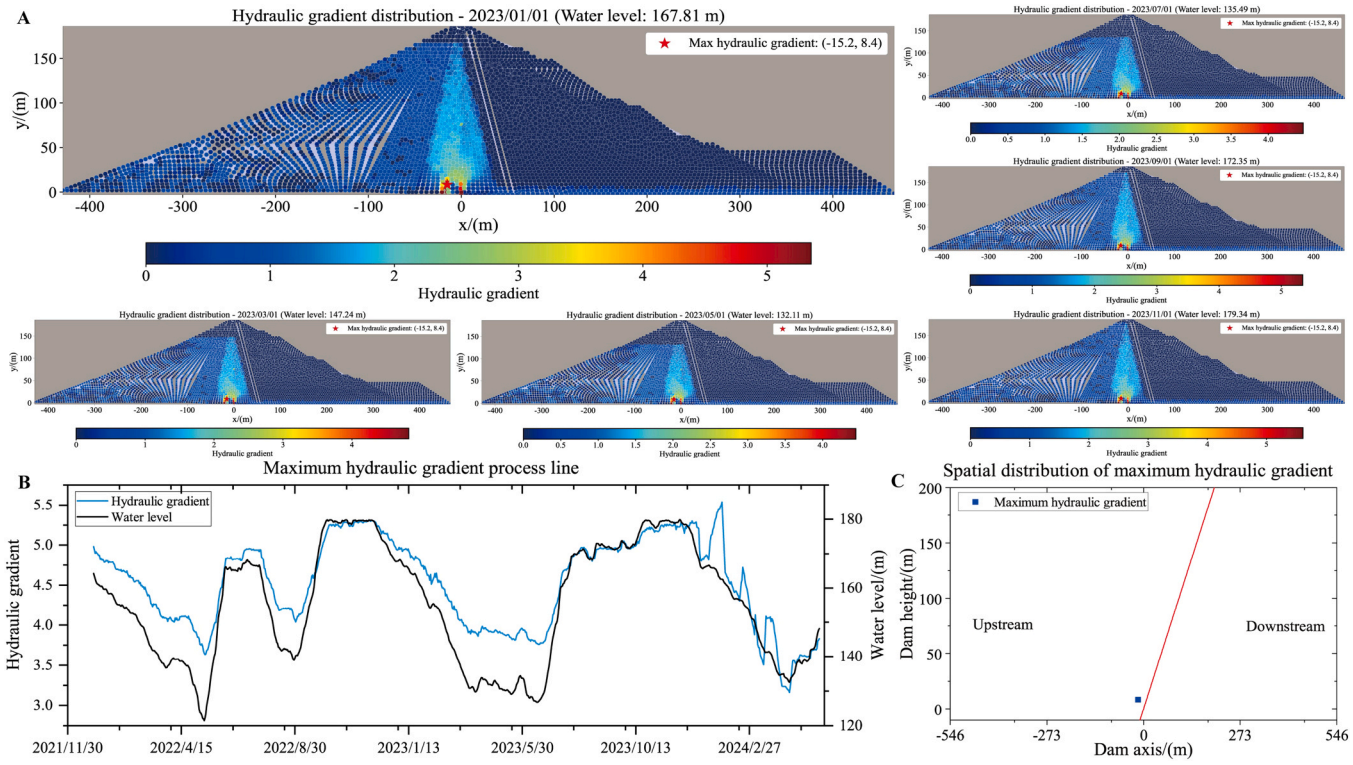

Fig. 8. Evaluation results of hydraulic gradient in the PB Dam digital twin.

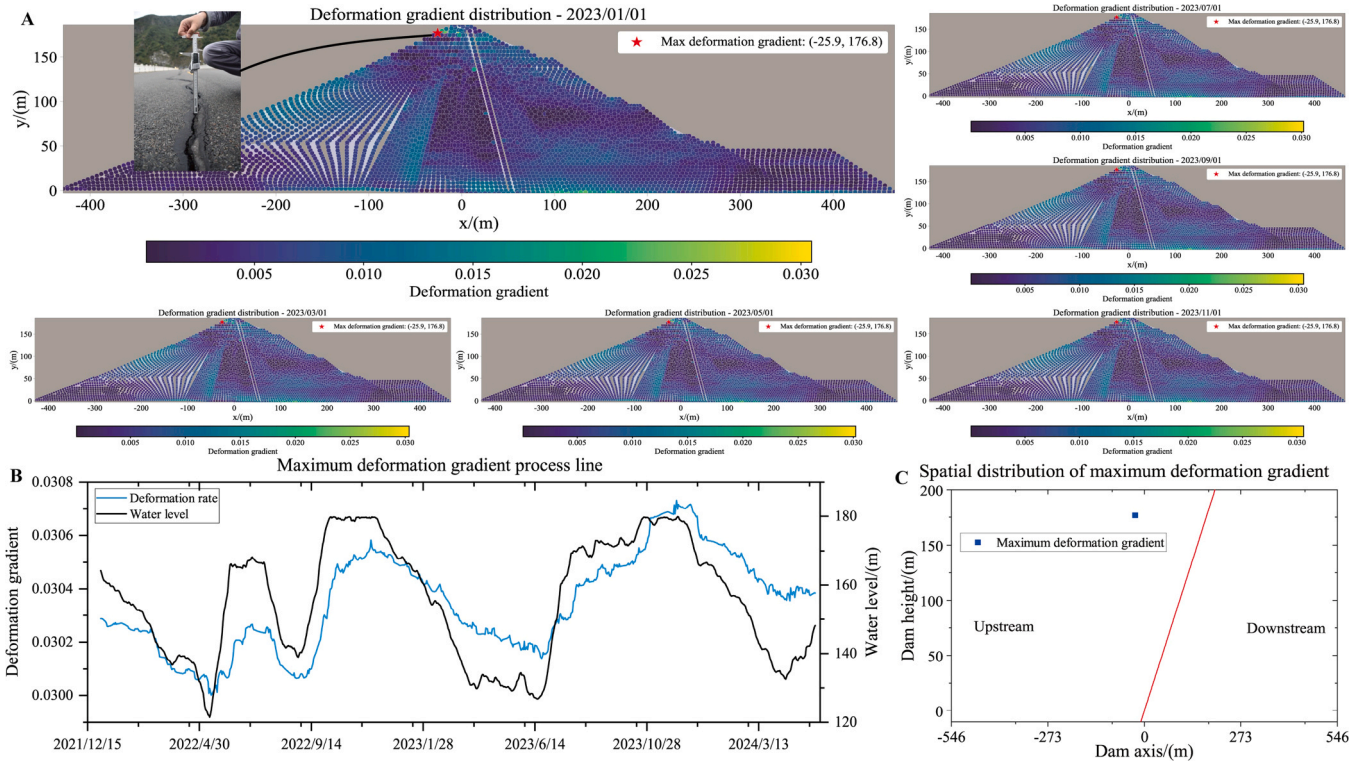

Fig. 9. Evaluation results of deformation gradient in the PB Dam digital twin.

actual operational conditions [57]. In addition, significant differential deformation is also observed in the downstream rockfill zone near the dam foundation, which is closely related to the presence of thick overlying layers. Fig. 9**B** presents the temporal evolution of the maximum deformation gradient, showing a certain coupling with water level fluctuations, with a maximum value of approximately 3.1%. The control of differential deformation mainly relies on empirical estimates. Overall, the deformation patterns predicted by the twin model are consistent with the typical deformation characteristics of high-core rockfill dams, effectively reflecting the development of cracking failure and providing

a reliable basis for structural safety assessment.

To further investigate the parameter sensitivity of the proposed method, a series of tests was conducted. When the Z-score threshold was fixed at $\tau = 2.5$, the number of nearest neighboring points was set to $k = 10$, 15, 20, and 25. Conversely, when the number of nearest neighboring points was fixed at $k = 20$, the Z-score threshold was varied as $\tau = 1$, 1.5, 2.5, and 4. The corresponding maximum deformation gradient evolution curves are shown in Fig. 10. The results indicate that the maximum deformation point identified under all parameter combinations is located at the same position, demonstrating the good robustness of the proposed method. Although the evolution trends of the curves are generally consistent, differences in the numerical values can be observed. The variations are more pronounced when the number of nearest neighboring points changes, whereas the influence of the Z-score threshold is relatively small. This phenomenon is mainly attributed to the influence of the computational neighborhood size, which is related to the finite element mesh density and element spacing. These factors affect the selection of the deformation gradient-based failure indicator, while the overall evolution pattern remains consistent across different parameter settings.

As shown in Fig. 11, the evaluation results of the effective stress in the core wall of the PB Dam digital twin are presented. The results indicate that the effective stress is relatively low on the upstream side of the core wall and higher on the downstream side, primarily influenced by the seepage pressure distribution. The minimum effective stress points are mainly located on the upstream side near the top of the core wall (Fig. 11**D**), where hydraulic fracturing is more likely to occur. Fig. 11**B** shows the temporal variation of the minimum effective stress. Due to its location at the dam crest, the coupling with the water level is relatively weak. Fig. 11**C** also presents the temporal evolution of the maximum effective stress, which is mainly concentrated at the downstream base of the core wall and exhibits a strong coupling with water level fluctuations. Overall, the effective stress distribution and its dynamic evolution predicted by the digital twin model are consistent with the typical characteristics of high-core rockfill dams and effectively reflect the development trend of hydraulic fracturing in the core wall zone.

As shown in Fig. 12, the evaluation results of the deformation rate of the PB Dam digital twin are presented. The results indicate that the deformation rate is relatively high in the mid-section of the upstream rockfill zone, reaching approximately 0.01 mm/day, while the downstream rockfill zone exhibits smaller deformation rates, mainly influenced by the seepage process. The maximum deformation rates are primarily distributed in the central region of the upstream rockfill zone (Fig. 12**C**), with a relatively scattered spatial pattern, suggesting that this area is more susceptible to locally induced sliding triggered by seepage. Fig. 12**B** illustrates the temporal evolution of the maximum deformation rate. Due to the randomness and spatial dispersion of the peak locations, their coupling with water level variation is relatively weak. Nevertheless, the overall deformation-rate response of the PB Dam digital twin still shows a clear correlation with water level fluctuations. In general, the spatial distribution and dynamic evolution of deformation rates predicted by the digital twin model align well with the typical deformation characteristics of high-core rockfill dams and can accurately reflect the development trend of dam deformation rates.

As shown in Fig. 13, the evaluation results of slope stability in the PB Dam digital twin are presented. The results indicate that the upstream slope exhibits relatively low stability factors (with a minimum of approximately 1.4), while the downstream slope shows higher stability, largely influenced by the distribution of the seepage field. The most critical slip surfaces are all located on the upstream side of the dam. According to the minimum safety factor of 1.3 recommended by relevant guidelines, this region remains within the acceptable safety threshold; however, its overall stability is relatively weak, suggesting a moderate landslide risk. Fig. 13**B** illustrates the temporal evolution of the minimum slope stability factor. A strong coupling with water level can be observed, accompanied by pronounced fluctuations, indicating that hydraulic loading and seepage processes exert a significant influence on slope stability. Overall, the spatial distribution and dynamic evolution of slope stability factors predicted by the digital twin model align well with the typical characteristics of high-core rockfill dams and can accurately capture the changing trend of slope stability.

4.3. Intelligent decision-making results

To evaluate the effectiveness of the proposed method, the digital twin results of hydraulic gradient, deformation gradient, effective stress, deformation rate, and slope stability generated by the fault diagnosis from 31 December 2021–23 April 2024 are input into the intelligent decision-making for comprehensive analysis. The data semanticization engine converts the twin-field outputs into high-level semantic information, and its diagnostic results are shown in the Fig. 14. The dam-specific LLM then performs reasoning and diagnosis based on the semanticized information, and its final outputs are also illustrated in Fig. 15.

In the structural safety assessment, explicitly defined diagnostic thresholds are adopted to support reliable and automated identification of potential failure modes in earth-rock dams. The selection of thresholds is based on a combination of engineering design codes and data-driven statistical criteria. For hydraulic gradients, the allowable design values are used as control thresholds (6.0 for the core wall and 10.0 for the concrete cutoff wall), consistent with dam design specifications and seepage control principles. Deformation-related indicators, including deformation gradients and deformation rates, are evaluated using a data-driven three-sigma (3σ) criterion derived from long-term monitoring data. This approach enables objective identification of abnormal

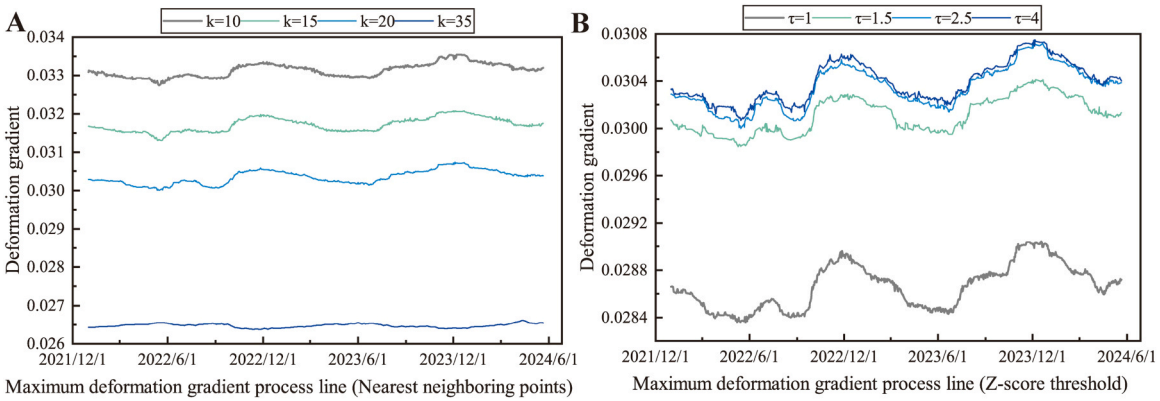

Fig. 10. Comparison of the influence of hyperparameters in the gradient calculation model on deformation gradient estimation.

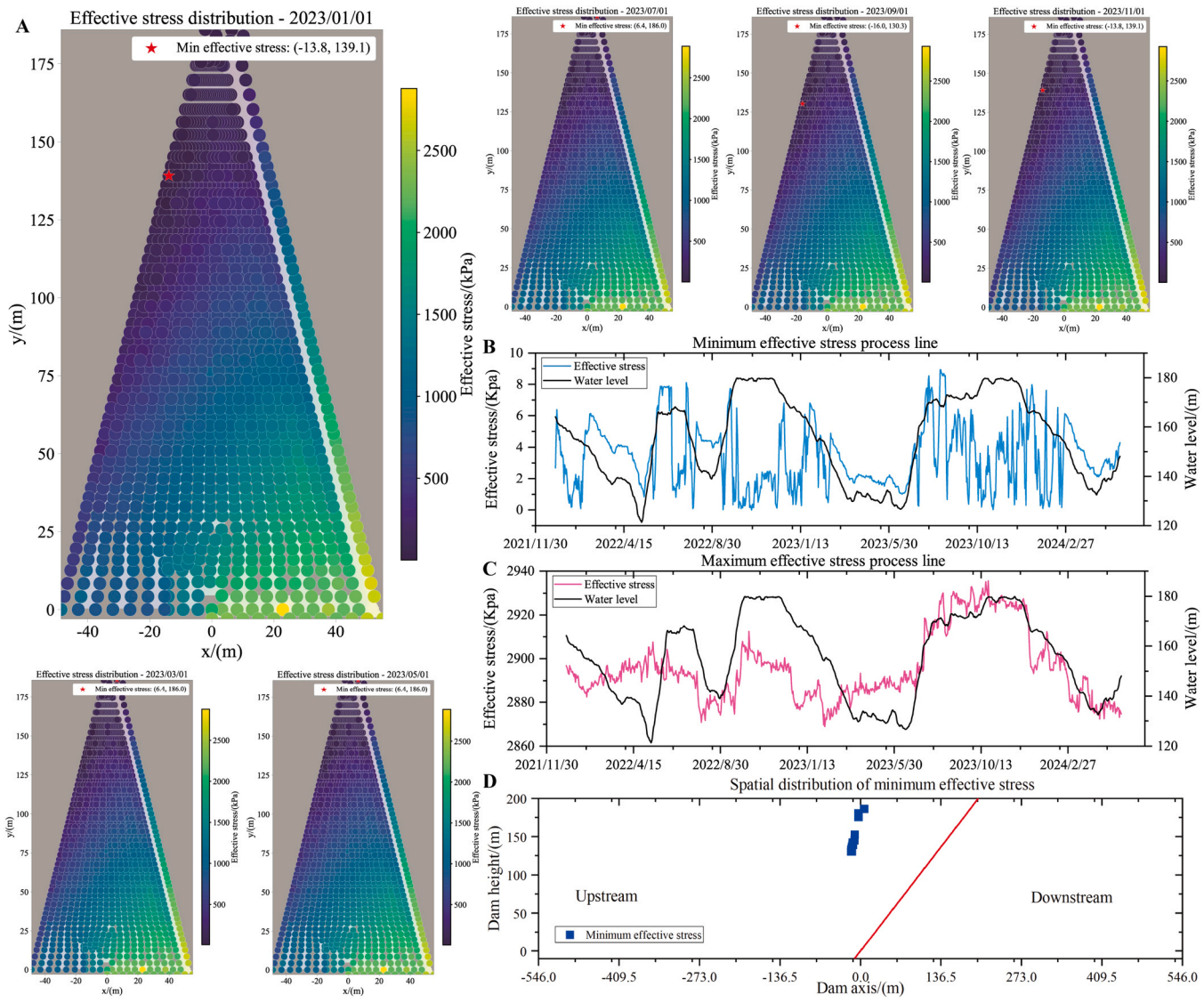

Fig. 11. Evaluation results of effective stress in the PB Dam digital twin.

structural responses by distinguishing statistically significant deviations from normal operational variability. Effective stress is assessed using tensile stress as the control indicator, as the occurrence of tensile stress in earth-rock dam materials indicates a potential loss of structural integrity and is therefore treated as a critical warning condition. Slope stability is evaluated based on the recommended safety factor of 1.3 specified in relevant engineering standards (NB/T 10872–2021), with values below this threshold indicating insufficient stability reserves. Based on the above criteria, the real-time operating condition of PB Dam is assessed. All monitored indicators remain within their respective threshold ranges, indicating that the dam is currently in an overall safe state. This diagnostic conclusion is consistent with observed monitoring data trends and actual operational conditions.

In the structural safety assessment, explicitly defined diagnostic thresholds are employed to enable reliable and automated identification of potential failure modes in earth-rock dams. These thresholds are determined based on national engineering design codes, empirical safety criteria, and long-term statistical analyses of monitoring data. For hydraulic gradients, the allowable limits specified in seepage control design standards are adopted, with threshold values of 6.0 for the core wall and 10.0 for the concrete cutoff wall. These criteria are widely applied in seepage safety evaluation to prevent internal erosion and hydraulic instability during long-term operation. Deformation-related indicators, including deformation gradients and deformation rates, are evaluated using a data-driven three-sigma (3σ) criterion derived from historical monitoring data under normal operating conditions. This approach facilitates objective anomaly detection by distinguishing abnormal structural responses from natural fluctuations in monitoring measurements. Effective stress is assessed using tensile stress as the primary control indicator, since the occurrence of tensile stress in earth-rock dam materials may indicate crack initiation or localized structural deterioration. Accordingly, tensile stress is regarded as an important warning signal in the fault diagnosis process. Slope stability is evaluated according to the minimum safety factor of 1.3 recommended in the engineering standard NB/T 10872–2021. Values below this threshold indicate insufficient stability reserves and potential slope instability risks. Based on these criteria, the real-time operating condition of PB Dam is comprehensively evaluated. All monitored indicators remain within their respective allowable ranges, indicating that the dam is currently operating in a safe condition. Furthermore, the diagnostic results are consistent with long-term monitoring trends and field operational observations.

To further verify the diagnostic capability of the proposed method, the thresholds are adjusted to conduct predictive simulations of different potential failure modes. The outputs of the data semanticization engine, corresponding to seepage failure, slope instability, crack development,

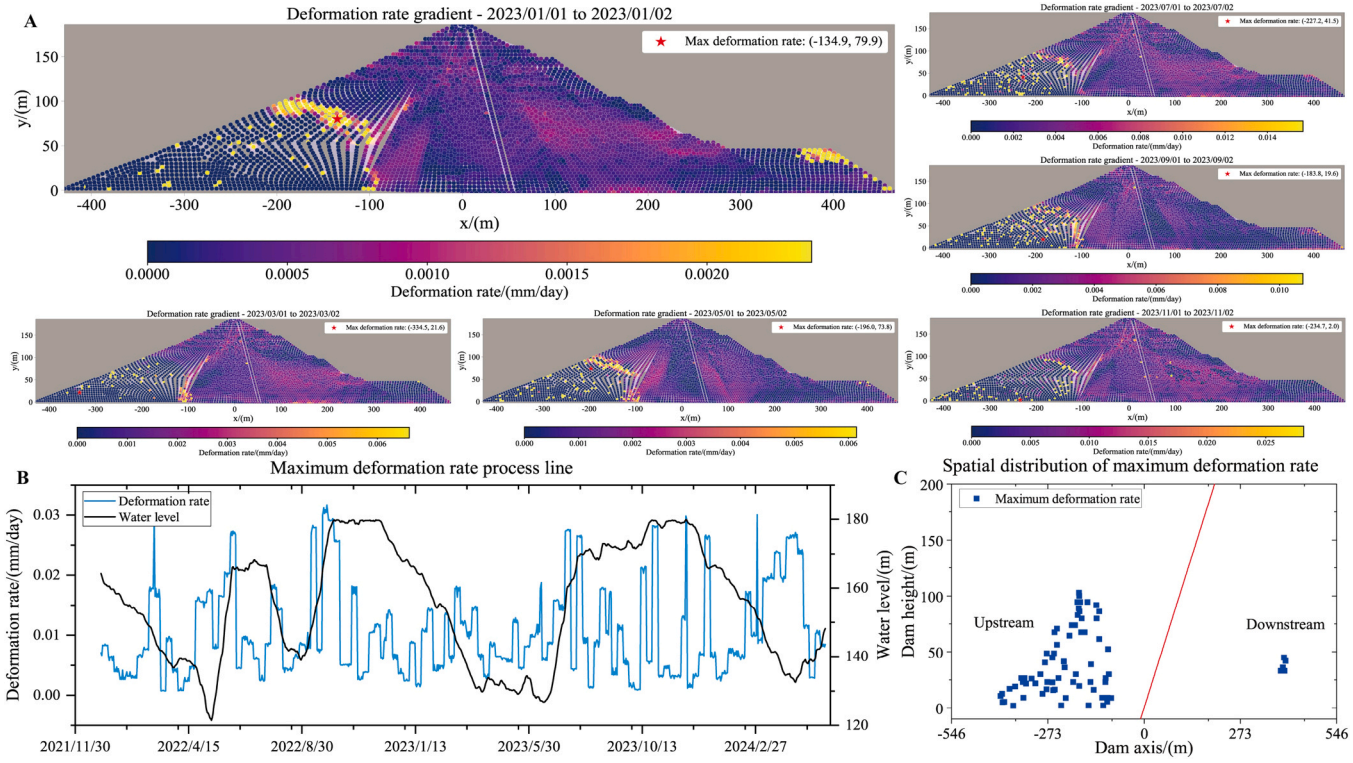

Fig. 12. Evaluation results of deformation rate in the PB Dam digital twin.

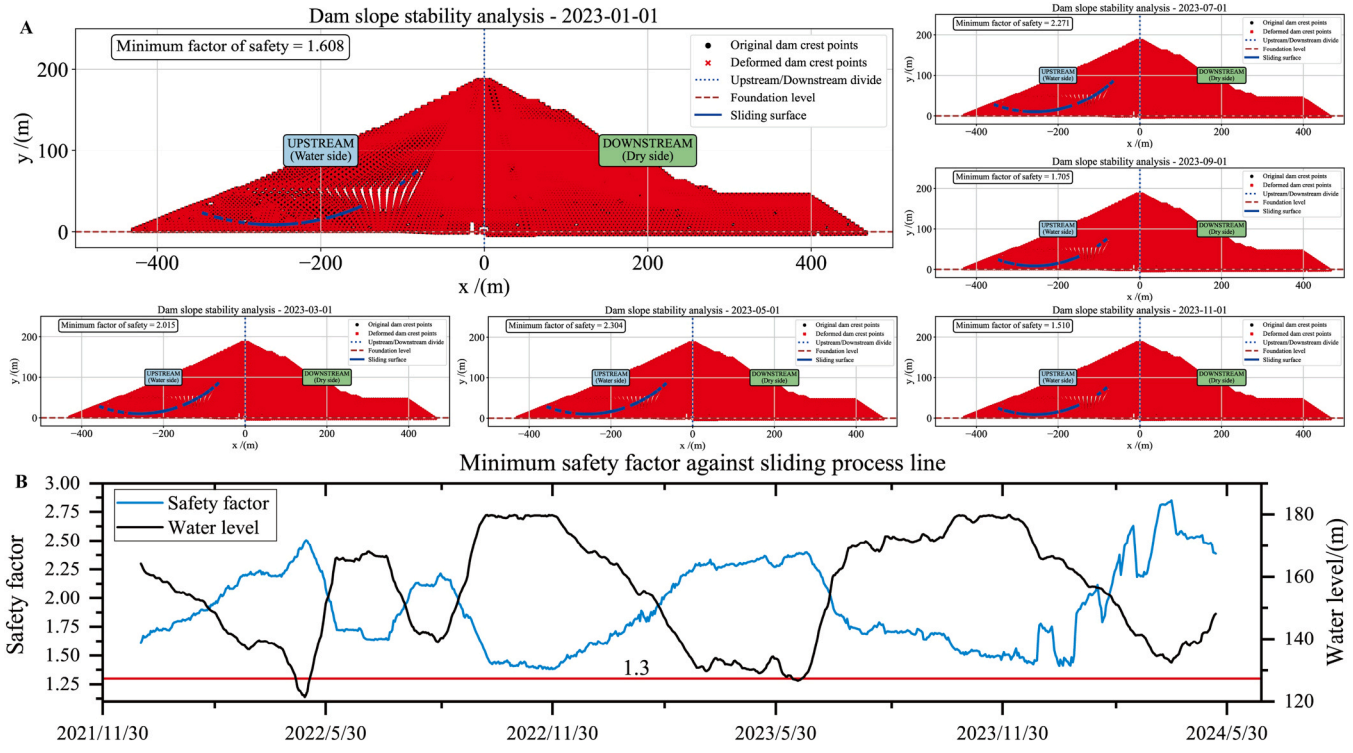

Fig. 13. Evaluation results of slope stability in the PB Dam digital twin.

and landslide modes, are shown in Fig. 14**A**, Fig. 14**B**, Fig. 14**C**, and Fig. 14**D**. The results show that the data semanticization engine effectively integrates multisource information from the deformation, seepage, and stress twin fields and, with the support of dam structural mechanics, transforms this information into natural-language descriptions that are aligned with engineering semantics. These findings demonstrate that the proposed reasoning framework provides strong capability for structural failure identification and intelligent diagnosis.

Fig. 15 presents the highly abstract natural-language information generated by the data semanticization engine and the reasoning results

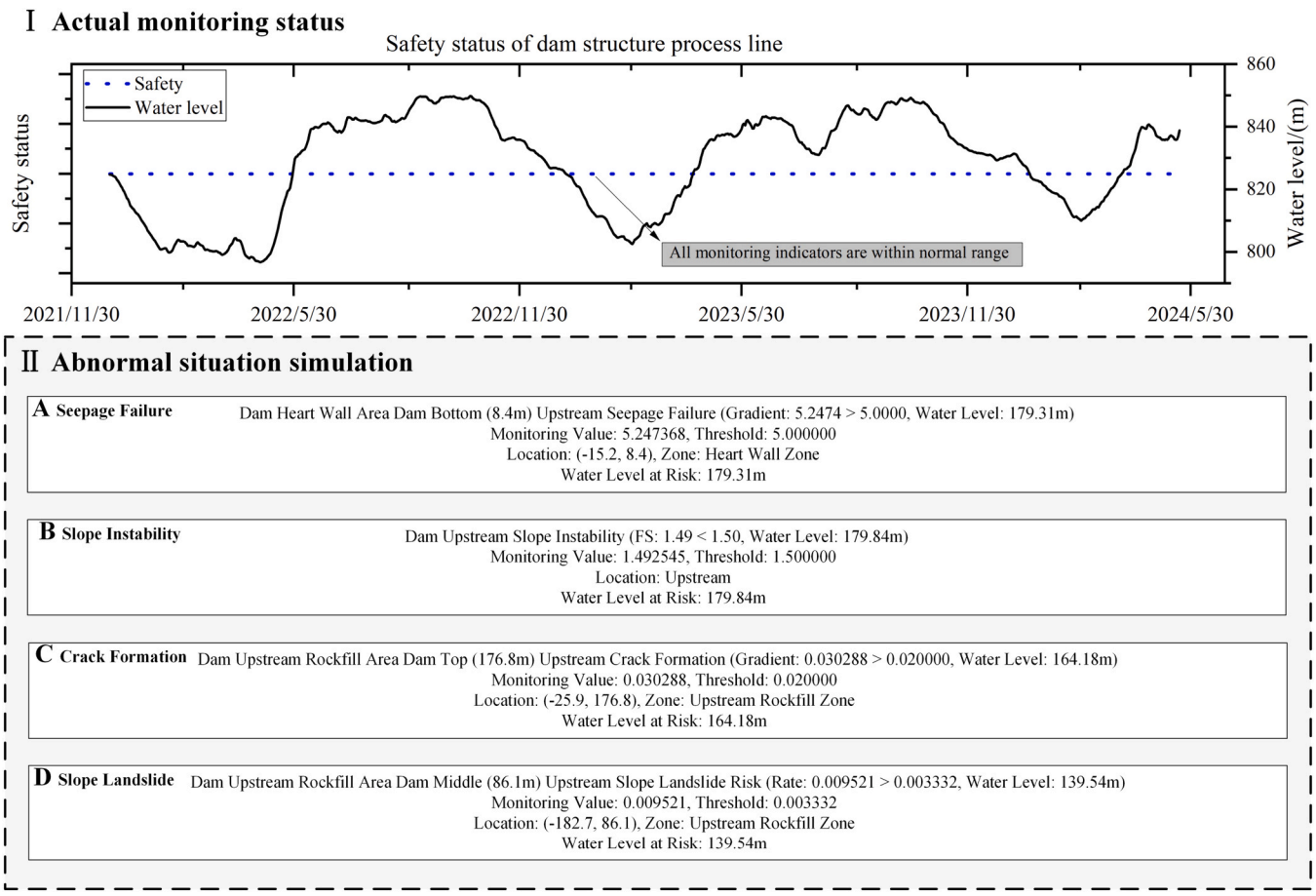

Fig. 14. Data semanticization engine diagnostic outputs and typical fault simulation.

produced by the dam-specific LLM. To evaluate the effectiveness of the proposed method, the predictive results of seepage failure were input separately into commonly used LLMs, including the dam-specific LLM, DeepSeek-V3.1, and ChatGPT (GPT−5.1 Thinking), for comparative analysis. The evaluation criteria were defined based on engineering standards and practical implementability, including: (1) terminological accuracy according to NB/T 10872–2021 “Design specification for rolled earth-rock fill dams”; (2) compliance with monitoring and safety assessment requirements in SLT551–2024 “Technical specification for earth-rock fill dam safety monitoring”; and (3) feasibility and operability of the proposed mitigation measures in real-world dam engineering practice.

The results indicate that the DeepSeek model, fine-tuned with a comprehensive dam engineering knowledge base, demonstrates superior performance. Its reasoning on failure causes and recommended mitigation measures aligns closely with both engineering practice and regulatory requirements. For instance, the term “deep overburden layer” corresponds accurately to the actual structure of PB Dam, while the “junction of the concrete cutoff wall and core wall” precisely identifies potential damage locations. Recommendations such as “enhanced monitoring” and “digital twin measures” satisfy the relevant requirements of SLT551–2024. Further comparison shows that the engineering measures generated by DeepSeek outperform those of ChatGPT in terms of operability, effectiveness, and real-time applicability. Specifically, DeepSeek distinguishes between short-term emergency responses and long-term remedial measures, prioritizing monitoring enhancement, early warning, and digital twin-based evaluation for immediate risk control. In contrast, ChatGPT tends to recommend long-term engineering interventions, such as downstream drainage improvement or grouting, without explicitly emphasizing monitoring-based measures that can be implemented in real time. This advantage is attributed to the integration of extensive Chinese dam engineering standards, technical manuals, and relevant literature into DeepSeek’s training. Overall, fine-tuning DeepSeek with a dam-specific knowledge base significantly enhances the LLM’s ability to interpret the outputs of the data semanticization engine, resulting in higher logical consistency, practical applicability, and engineering effectiveness.

5. Discussions

The hybrid framework integrating digital twin and LLM technologies proposed in this study provides a practical solution for SHM and operational management of high earth-rock dams. By combining multi-physics digital twin prediction with knowledge-driven reasoning, the framework improves the analysis capability for complex monitoring data and enhances the interpretability of diagnostic results. In engineering practice, the framework can continuously integrate monitoring data with digital twin simulation outputs to identify abnormal structural behavior, generate interpretable diagnostic reports, and support routine dam safety assessment and maintenance management. During emergency conditions, such as abnormal seepage or deformation evolution, the system can rapidly infer potential failure modes and provide targeted mitigation recommendations, thereby assisting engineers in timely operational decision-making. Owing to its modular architecture and compatibility with monitoring systems and numerical simulation models, the proposed framework also shows potential for broader application to other large hydraulic infrastructures and intelligent infrastructure monitoring scenarios.

Digital twin methods, as advanced monitoring tools in dam engineering, can, through the combination of finite element analysis and

Fig. 15. Comparison of reasoning results from different large language models.

neural network-based transfer learning, accurately reproduce dam deformation, seepage, and stress behaviors [13,28]. However, there remains a significant gap in the analysis and intelligent diagnosis of twin outputs [14,48]. To address this issue, the study establishes a fault-type diagnosis model for high earth-rock dams based on digital twin outputs. This model, relying on gradient computation and slope stability analysis, develops a robust gradient estimation method and is capable of performing high-precision calculations in real time by integrating deformation twin, seepage twin, stress twin, and time-varying parameter information, thereby significantly improving the real-time performance and effectiveness of fault diagnosis. Nevertheless, diagnostic results in conventional methods still heavily depend on human expert operation and judgment. To overcome this limitation, the study introduces, for the first time, the conversion of multisource data into natural-language information and implements intelligent suggestions and human-machine interaction through a reasoning model. Furthermore, by constructing a dam engineering knowledge base and fine-tuning the LLM, the reasoning and predictive outputs of the model are constrained, making them more reliable and controllable in engineering applications.

Although the proposed framework has been validated and analyzed through a fully monitored high-core rockfill dam case study, several aspects still require further investigation and optimization. For instance, the accuracy of digital twin modelling may be affected under less comprehensive monitoring layouts or for different dam types, such as concrete face rockfill dams and concrete dams. While the overall framework remains applicable to concrete dams, the corresponding structural failure assessment methods and safety evaluation indicators require further refinement to better capture the distinct mechanical behavior and damage characteristics of concrete structures.

6. Conclusions

This study proposes a hybrid framework for intelligent structural health monitoring of high earth-rock dams by integrating multi-physics digital twins with LLMs. Unlike conventional digital twin approaches that mainly focus on state prediction, the proposed framework further incorporates structural fault diagnosis, semantic interpretation, and intelligent engineering decision-making, thereby establishing a closed-loop workflow from multi-source monitoring perception to fault reasoning and recommendation generation. The framework consists of three core components, including multi-physics digital twin modelling, intelligent structural fault diagnosis, and LLM-assisted intelligent decision-making, and its effectiveness is validated through a fully monitored high core-wall rockfill dam case. The main conclusions are summarized as follows:

- The proposed method enables high-precision full-domain reconstruction of deformation, seepage, and stress fields using limited monitoring information and numerical simulation data. By integrating transfer learning, graph attention networks, and physics-

based constraints, the digital twin model effectively captures the spatial evolution characteristics of multi-physics fields and provides reliable support for real-time structural safety assessment.

- The proposed fault diagnosis method successfully identifies typical failure modes of earth-rock dams, including seepage failure, hydraulic fracturing, cracking, and landslide risk. In particular, the robust gradient estimation strategy and the real-time slope stability evaluation model incorporating time-varying material parameters significantly improve the stability and interpretability of structural diagnosis under complex hydro-mechanical conditions.
- A data semanticization engine is developed to transform digital diagnostic results into engineering-oriented natural-language descriptions by integrating structural mechanics knowledge and monitoring information. Combined with the dam-specific LLM, the framework achieves intelligent fault reasoning, root-cause analysis, and mitigation recommendation generation, demonstrating strong potential for human-machine interaction and intelligent operational management of hydraulic infrastructures.

Nevertheless, the proposed framework still has several limitations. The current digital twin model remains dependent on the quality and density of monitoring data, while uncertainties associated with material parameter evolution and cross-domain discrepancies between numerical simulations and field observations may affect long-term prediction accuracy. In addition, the intelligent decision-making module is primarily based on offline fine-tuning and engineering knowledge retrieval, and its real-time reasoning robustness under complex operational scenarios still requires further validation. Future work will focus on uncertainty-aware digital twin updating, BIM-FEM integrated information modelling, information-entropy-based safety evaluation, and reinforcement learning-assisted LLM optimization, with the aim of improving the generalization capability, interpretability, and practical applicability of intelligent SHM systems for large-scale earth-rock dams.

CRediT authorship contribution statement

Jichen Tian: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Chen Chen:** Writing – review & editing, Supervision, Resources. **Yanling Li:** Writing – review & editing, Supervision, Data curation. **huang Huibao:** Writing – review & editing, Formal analysis, Data curation. **Jiankang Chen:** Writing – review & editing, Resources, Investigation.

Declaration of Competing Interest

The author(s) declared no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

Acknowledgment

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article. This work was supported by the State Key Program of the National Natural Science Foundation of China (No.52130904) and the National Natural Science Foundation of China (No.42277181).

Data Availability

Data will be made available on request.

References

[1] He JX, Li H, Li J. Permeability evolution and prediction method of hydraulic asphalt concrete under confining pressure effect. Constr Build Mater 2025;492. https://doi.org/10.1016/j.conbuildmat.2025.142853.
[2] Yao Q, Wu Y, He J, Qi SC, Li HT. A New Method for the Compaction Quality Inspection of High Rockfill Dams Based on 3D Laser Scanning Technology. Struct Control & Health Monit 2024;2024. https://doi.org/10.1155/2024/6662678.
[3] Yang FC, Liu HL, Xiao Y, Shi JQ, Peng Y, Ding XM, et al. Mechanical degradation and particle breakage of sandstone rockfill material under wetting-drying cycles. Acta Geotech 2025;20:2237–55. https://doi.org/10.1007/s11440-025-02545-7.
[4] Tian JC, Chen C, Lu X, Li YL, Chen JK. Deep transfer learning-based time-varying model for deformation monitoring of high earth-rock dams. Eng Appl Artif Intell 2024;138. https://doi.org/10.1016/j.engappai.2024.109310.
[5] Lan XD, Li YB, Zou S, Liang YH. Investigation of the splitting tensile behavior of rock-filled concrete based on mesoscopic modelling: Considering the effects of rock strength, bearing strip width, and rockfill ratio. Structures 2025;82. https://doi.org/10.1016/j.istruc.2025.110432.
[6] Bayraktar A, Ventura CE, Yang TY, Hökelekli E, Tas Y. Observed Damage Behavior of Earth Dams During the 2023 Kahramanmaraş, Turkey, Earthquakes. Geotech Geol Eng 2024;42:7025–40. https://doi.org/10.1007/s10706-024-02911-4.
[7] Liu HC, Su HZ, Kookalani S, Cao MS, Wu Lie YM. WY. Attention-enhanced spatiotemporal deep learning-based automatic warning model with uncertainty estimation in dam safety. Struct Health Monit- Int J 2025. https://doi.org/10.1177/14759217251358537.
[8] Flah M, Nunez I, Ben Chaabene W, Nehdi ML. Machine Learning Algorithms in Civil Structural Health Monitoring: A Systematic Review. Arch Comput Methods Eng 2021;28:2621–43. https://doi.org/10.1007/s11831-020-09471-9.
[9] Alshuhail A, Mengash HA, Alanazi MH, Saeed MK, Ghaleb M, Duhayyim Al, et al. A. TinyML-enabled structural health monitoring for real-time anomaly detection in civil infrastructure. Alex Eng J 2025;129:1340–8. https://doi.org/10.1016/j.aej.2025.08.046.
[10] Cha YJ, Ali R, Lewis J, Buyukozturk O. Deep learning-based structural health monitoring. Autom Constr 2024;161. https://doi.org/10.1016/j.autcon.2024.105328.
[11] Li ZC, Khailah EY, Liu XY, Liang JM. Exploring Purpose-Driven Methods and a Multifaceted Approach in Dam Health Monitoring Data Utilization. Buildings 2025;15. https://doi.org/10.3390/buildings15152803.
[12] Tian JC, Chen JK, Wang JY, Huang HB, Li YL. Real-time classification model for anomalous sensor data in dam safety monitoring based on convolutional neural networks and transfer learning. Expert Syst Appl 2026;296. https://doi.org/10.1016/j.eswa.2025.129135.
[13] Tian JC, Yu RJ, Chen JK, Chen C, Li YL, Sun XJ. Haung HB. Digital twin model for analyzing deformation and seepage in high earth-rock dams. Autom Constr 2025;173. https://doi.org/10.1016/j.autcon.2025.106079.
[14] Tian JC, Yu RJ, Chen C, Li Y, Huang HB, Chen JK. Cross-domain adaptive digital twin modelling for dam structures using graph attention networks. Adv Eng Inform 2026;70:104209. https://doi.org/10.1016/j.aei.2025.104209.
[15] Wang R, Cui F, Zhao XX, Dong GQ, Cheng Q, Zhang GX. Pipeline structure and leakage detection method of full polarimetric ground penetrating radar based on model and eigenvalue decomposition. Autom Constr 2025;178. https://doi.org/10.1016/j.autcon.2025.106391.
[16] Rao AS, Radanovic M, Liu YG, Hu SB, Fang YH, Khoshelham K, et al. Tuan N. Real-time monitoring of construction sites: Sensors, methods, and applications. Autom Constr 2022;136. https://doi.org/10.1016/j.autcon.2021.104099.
[17] Li B, Yang J, Hu DX. Dam monitoring data analysis methods: A literature review. Struct Control & Health Monit 2020;27. https://doi.org/10.1002/stc.2501.
[18] Liu XY, Li ZC, Sun LS, Khailah EY, Wang JJ, Lu WG. A critical review of statistical model of dam monitoring data. J Build Eng 2023;80. https://doi.org/10.1016/j.jobe.2023.108106.
[19] Huang HR, Pang Q, Yan JH, Chen SY, Wang SJ, Wu YX, et al. A seepage prediction method for earth-rock dams integrating surrogate modelling with hysteresis effect functions. J Civil Struct Health Monit 2025;15:3695–711. https://doi.org/10.1007/s13349-025-01011-y.
[20] Zhang PT, Chen C, Zhang B, Pei L, Lu X, Chen JK, et al. A dynamic assurance rate methodology for analyzing abutments stability of arch dams based on deformation monitoring data. Struct Health Monit- Int J 2025. https://doi.org/10.1177/14759217251387037.
[21] Abdikan S, Coskun S, Narin OG, Ozdemir EG, Bayik C, Sanli FB. Deep-learning-based interferometric synthetic aperture radar time-series analysis for the monitoring and prediction of dam safety. Struct Health Monit- Int J 2025. https://doi.org/10.1177/14759217251381157.
[22] Liu MK, Qi YN, Su HZ. Deformation prediction model for concrete dams considering the effect of solar radiation. Adv Eng Inform 2025;65. https://doi.org/10.1016/j.aei.2025.103252.
[23] Li MC, Li MH, Ren QB, Li H, Song LG. DRLSTM: A dual-stage deep learning approach driven by raw monitoring data for dam displacement prediction. Adv Eng Inform 2022;51. https://doi.org/10.1016/j.aei.2021.101510.
[24] Tian JC, Luo YH, Huang HB, Chen JK, Li YL. Rapid postearthquake modelling method for deformation monitoring models of high arch dams based on metalearning and graph attention. Adv Eng Inform 2024;62. https://doi.org/10.1016/j.aei.2024.102925.
[25] Tian JC, Li YL, Luo YH, Zhang H, Lu X. Multisource information fusion model for deformation safety monitoring of earth and rock dams based on deep graph feature fusion. Struct Health Monit- Int J 2025;24:925–40. https://doi.org/10.1177/14759217241244549.
[26] Faraji A, Arya SH, Ghasemi E, Abagheri HA, Perera S. Exploring the metaverse: integrating emerging technologies for Industry 5.0 advancement. Smart Sustain Built Environ 2025. https://doi.org/10.1108/sasbe-04-2025-0169.
[27] Ligarda-Samanez CA, Huamán-Carrión ML, Cabel-Moscoso DJ, Sáenz DMM, Hernandez JAM, Garcia-Espinoza AJ, et al. Technological Innovations in

Sustainable Civil Engineering: Advanced Materials, Resilient Design, and Digital Tools. Sustainability 2025;17. https://doi.org/10.3390/su17198741.
[28] Tian JC, Chen C, Zhang LM, Chen JK, Huang HB, Zhang PT. A digital twin method for real-time analysis of structural deformation and failure for high arch dams. Eng Struct 2025;345. https://doi.org/10.1016/j.engstruct.2025.121426.
[29] Rinaldi C, Bendarma A, Potenza F, et al. Physics-informed computational framework for assessing the seismic behavior of the Prince Moulay Abdellah Dam. Results Eng 2025;28. https://doi.org/10.1016/j.rineng.2025.106914.
[30] Anari MK, Mirhosseini SM, Lajevardi SH, et al. Effect of foundation stiffness on the fragility curves of a concrete gravity dam under far-field ground motions. Results Eng 2024;24. https://doi.org/10.1016/j.rineng.2024.102962.
[31] Beiranvand B, Rajaee T, Komasi M. Spatiotemporal clustering of dam settlement monitoring using instrumentation data (case study: Eyvashan Earth Dam). Results Eng 2024;22. https://doi.org/10.1016/j.rineng.2024.102014.
[32] Xu Z, Rao ZZ, Gan VJL, et al. Developing an Extended IFC Data Schema and Mesh Generation Framework for Finite Element Modelling. Adv Civil Eng 2019;2019. https://doi.org/10.1155/2019/1434093.
[33] Zhu HM, Huang MQ, Ji PX, Xiao F, Zhang QB. Transforming the maintenance of underground infrastructure through Digital Twins: State of the art and outlook. Tunn Undergr Space Technol 2025;161. https://doi.org/10.1016/j.tust.2025.106508.
[34] Tuhaise VV, Tah JHM, Abanda FH. Technologies for digital twin applications in construction. Autom Constr 2023;152. https://doi.org/10.1016/j.autcon.2023.104931.
[35] Hu JQ, Xiao H, Ye ZH, Luo NZ, Zhou MH. Research and Prospects of Digital Twin-Based Fault Diagnosis of Electric Machines. Sensors 2025;25. https://doi.org/10.3390/s25082625.
[36] Kolate VD, Chaudhary MK, Mane S, Devadhe M. Digital twin-enabled model predictive control in additive manufacturing: critical review, research challenges, and future directions. Mater Manuf Process 2026;41:1–17. https://doi.org/10.1080/10426914.2025.2586498.
[37] Yang YL, Zhu Y, Cai CS. Research progress and prospect of digital twin in bridge engineering. Adv Struct Eng 2024;27:333–52. https://doi.org/10.1177/13694332231218764.
[38] Lai XA, Kan ZY, Sun W, Song XG, Tian BM, Yuan TF. Digital twin-based non-destructive testing for structural health monitoring of bridges. Nondestruct Test Eval 2024;39:57–74. https://doi.org/10.1080/10589759.2023.2239434.
[39] Li XC, Ye LJ, Bian XC. Sequential data assimilation for digital twin modelling of shield tunnel structure-soil interaction systems. Tunn Undergr Space Technol 2026;168. https://doi.org/10.1016/j.tust.2025.107168.
[40] Li HY, Xu YL, Liao HL, Huang L, Wang Q. Digital twin-driven method for determining wind force coefficients of a bridge deck section. Eng Appl Comput Fluid Mech 2025;19. https://doi.org/10.1080/19942060.2025.2501389.
[41] Tang K, Xiao H, Wang DH, Chen G. Physics-informed spatiotemporal architecture in digital twin modelling for condition monitoring of aircraft engines. Adv Eng Inform 2026;69. https://doi.org/10.1016/j.aei.2025.104039.
[42] Zheng CM, Wong SK, Su X, Tang YQ, Nawaz A, Kassem M. Automating construction contract review using knowledge graph-enhanced large language models. Autom Constr 2025;175. https://doi.org/10.1016/j.autcon.2025.106179.
[43] Ao Y, Li SF, Duan HL. Artificial Intelligence-Aided Design (AIAD) for Structures and Engineering: A State-of-the-Art Review and Future Perspectives. Arch Comput Methods Eng 2025;32:4197–224. https://doi.org/10.1007/s11831-025-10264-1.
[44] Zhong YS, Goodfellow SD. Domain-specific language models pre-trained on construction management systems corpora. Autom Constr 2024;160. https://doi.org/10.1016/j.autcon.2024.105316.
[45] Jiang G, Chen JL. Efficient fine-tuning of large language models for automated building energy modelling in complex cases. Autom Constr 2025;175. https://doi.org/10.1016/j.autcon.2025.106223.
[46] Yang LY, Chen HY, Li Z, Ding X, Wu XD. Give us the Facts: Enhancing Large Language Models With Knowledge Graphs for Fact-Aware Language Modelling. Ieee Trans Knowl Data Eng 2024;36:3091–110. https://doi.org/10.1109/tkde.2024.3360454.
[47] Lin XS, Chen WQ, Zhou Z, Li JH, Zhao YM, Zhang XY. A five-dimensional digital twin framework driven by large language models-enhanced RL for CNC systems. Robot Comput-Integr Manuf 2025:95. https://doi.org/10.1016/j.rcim.2025.103009.
[48] Luo PH, Wen XF, Jiang YX, Parn E, Brilakis L. ChatTwin: Bridging the usability gap to digital twin adoption in infrastructure operations and maintenance with a Natural Language Interface. Autom Constr 2026;181. https://doi.org/10.1016/j.autcon.2025.106655.
[49] Banad YM, Sharif SS, Rezaei Z. Artificial intelligence and machine learning for smart grids: from foundational paradigms to emerging technologies with digital twin and large language model-driven intelligence. Energy Convers Manag-X 2025;28. https://doi.org/10.1016/j.ecmx.2025.101329.
[50] Guo FY, Wang CC, Sun J, et al. AN INTELLIGENT CONSTRUCTION SYSTEM BASED ON DIGITAL TWIN AND FOUNDATION MODEL OPTIMIZATION. J Civil Eng Manag 2026;32:374–93. https://doi.org/10.3846/jcem.2026.26154.
[51] Gao Y, Yiu TW, Shen XS, et al. Large language models in smart construction: a systematic review of implementation strategies, applications and future directions. Eng Constr Archit Manag 2026;33:159–81. https://doi.org/10.1108/ecam-10-2025-1668.
[52] Liu BR, Chen HN. BIMCoder: A Comprehensive Large Language Model Fusion Framework for Natural Language-Based BIM Information Retrieval. Appl Sci-Basel 2025;15. https://doi.org/10.3390/app15147647.
[53] Soy M. and Zhang J.S. GPT4-Vision Multimodal Model-Powered Query-Answering Chatbot for Bridge PDF Drawings. In: 2024 International Conference on Computing in Civil Engineering-I3CE Carnegie Mellon University, Pittsburgh, PA, Jul 28-31, 2024 2024, Computing in civil engineering 2024: Artificial intelligence, automation and robotics, and human-centered innovations, pp.505-513.
[54] Su HZ, Zhang S, Wen ZP, Li H. Prototype monitoring data-based analysis of time-varying material parameters of dams and their foundation with structural reinforcement (Article.) Eng Comput 2017;33:1027–43. https://doi.org/10.1007/s00366-017-0514-1.
[55] Fu W, Sheng X, Chen Q, Zhou J, Zheng YQ. SJ. Physics-based and data-driven long-term evolution of a landslide: From inversion to prediction (Article.) Eng Geol 2025;355. https://doi.org/10.1016/j.enggeo.2025.108252.
[56] Tian JC, Zhou W, Li YL, Chen C, Chen JK, Huang HB. A few-shot modelling method for early-stage safety monitoring of high earth-rock dams based on multi-source domain adaptation. Adv Eng Inform 2026;70:104194. https://doi.org/10.1016/j.aei.2025.104194.
[57] Zhang H, Jing YL, Chen JK, Gao ZL, Xu Y. Characteristics and causes of crest cracking on a high core-wall rockfill dam: A case study. Eng Geol 2022;297. https://doi.org/10.1016/j.enggeo.2021.106488.